10.8.16 Let $f : [0,1] \to \mathbb{R}$ be a continuous function with the property that $f(0) = 0$ and

$$\int_0^1 f(x) \sin \pi n x \, dx = 0$$

for all $n = 1, 2, 3, \ldots$. What can you conclude about the function f?

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Notes

[250]Exercise 10.2.2. This follows immediately from the inequalities

$$\liminf_k \left| \frac{a_{k+1}}{a_k} \right| \leq \liminf \sqrt[k]{|a_k|}$$

$$\leq \limsup_k \sqrt[k]{|a_k|} \leq \limsup_k \left| \frac{a_{k+1}}{a_k} \right|$$

that we obtained in Exercise 2.13.16.

[251]Exercise 10.3.3. Write out the Cauchy criterion for uniform convergence on $(-r, r)$ and deduce that the Cauchy criterion for uniform convergence on $[-r, r]$ must also hold.

[252]Exercise 10.4.4.

$$\int_0^1 \frac{1 - e^{-sx}}{s} \, ds = \sum_{k=1}^{\infty} (-1)^{k-1} \frac{1}{k(k!)} x^k.$$

[253]Exercise 10.5.4. It is clear that $f^{(k)}$ exists for all $x \neq 0$. For $x = 0$ verify the following assertions:

1. $f^{(k)}(x)$ is of the form $R(x)e^{-1/x^2}$ for $x \neq 0$, where R is a rational function.
2. Show that
$$\lim_{x \to 0} \frac{1}{x^n} e^{-1/x^2} = 0$$
for all $n = 1, 2, \ldots$.
3. Conclude that
$$\lim_{x \to 0} f^{(k)}(x) = 0$$
for all $k = 1, 2, \ldots$.
4. Conclude that
$$f^{(k)}(0) = 0$$
for all k.

[254]Exercise 10.6.2. Just use Theorem 10.32.

[255]Exercise 10.7.1. Just use Theorem 10.33.

[256]Exercise 10.8.5. Easy, really. Just substitute

$$u = x + t$$

in the integral

$$\int_0^{\pi} f(x+t) D_n(t) \, dt$$

and expand the terms $\cos(ku-kx)$ using standard trigonometric identities.

[257]Exercise 10.8.12. First obtain a polynomial q so that

$$|f(x)-q(x)|<\varepsilon/2.$$

Then find a polynomial p with rational coefficients so that

$$|p(x)-q(x)|<\varepsilon/2.$$

[258]Exercise 10.8.13. Try $f(x)=e^x$.

[259]Exercise 10.8.14. Try $f(x)=1/x$.

[260]Exercise 10.8.15. Show that f must be identically equal to zero. Use Theorem 10.37.

[261]Exercise 10.8.16. Define

$$G(t)=f(t/\pi)$$

for $t\in[0,\pi]$ and extend to $[-\pi,0]$ by

$$G(-t)=-G(t).$$

Consider the Fourier series of G and show that it contains only sin terms (no cosine terms). Show that f must be identically equal to zero. Use Theorem 10.36.

Chapter 11

THE EUCLIDEAN SPACES $\mathbb{R}^n$

In our study of analysis we have observed that the real number system $\mathbb{R}$ is more than just a collection of points. It has various algebraic, metric, and order structures that play a role in the development of the subject. The same is true of many other important sets of mathematical objects. Among the most important are the euclidean spaces $\mathbb{R}^n$, which we study in this chapter. The collection of elements of the set $\mathbb{R}^n$ is easy to describe: It is just the set of all n-tuples of real numbers

$$x = (x_1, \ldots, x_n).$$

We'll spend some time developing a natural algebraic structure (the space $\mathbb{R}^n$ forms a vector space) and a natural metric structure (there are natural notions of distance) that we'll exploit in this and the next chapter. (While we can also impose order structures on these spaces, such structures are not natural for our purposes and will not enter our discussions.)

11.1 The Algebraic Structure of $\mathbb{R}^n$

The space $\mathbb{R}^n$ consists of all n-tuples $x = (x_1, \ldots, x_n)$ of real numbers. That is, $\mathbb{R}^n$ is the cartesian product of n copies of $\mathbb{R}$:

$$\mathbb{R}^n = \mathbb{R} \times \mathbb{R} \times \cdots \times \mathbb{R} \quad (n \text{ factors}).$$

The members x of $\mathbb{R}^n$ are called the points of $\mathbb{R}^n$. They can also be viewed as *vectors*, and $\mathbb{R}$ as a field of *scalars*. We define addition of vectors in $\mathbb{R}^n$ as coordinate-wise addition: If $x = (x_1, \ldots, x_n)$ and $y = (y_1, \ldots, y_n)$, then

$$x + y = (x_1 + y_1, \ldots, x_n + y_n). \tag{1}$$

For $\alpha \in \mathbb{R}$ and $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we define

$$\alpha x = (\alpha x_1, \ldots, \alpha x_n). \tag{2}$$

We refer to such multiplication of a vector by a scalar as *scalar multiplication*.

With these notions of addition and scalar multiplication, $\mathbb{R}^n$ possesses the algebraic structure of a *vector space*. This means that the properties indicated in Theorem 11.1 are valid.

Theorem 11.1: *If addition and scalar multiplication on $\mathbb{R}^n$ are defined by (1) and (2), respectively, then the following are true for all x, y, $z \in \mathbb{R}^n$ and all α, $\beta \in \mathbb{R}$:*

$x+y$ *and* αx *are in* $\mathbb{R}^n$	*(closure of* $\mathbb{R}^n$ *under addition and scalar multiplication)*
$x+y=y+x$	*(commutative law)*
$(x+y)+z=x+(y+z)$	*(associative law)*
$x+0=x$	*where* $0=(0,\ldots,0)$
$\alpha(x+y)=\alpha x+\alpha y$	*(distributive*
$(\alpha+\beta)x=\alpha x+\beta x$	*laws)*
$\alpha(\beta x)=(\alpha\beta)x$	
$0x=0$	
$1x=x.$	

The proof of Theorem 11.1 is just a simple exercise of checking each statement. We leave it as Exercise 11.1.1. We also observe that by defining subtraction by

$$x-y=x+(-1)y,$$

then $x-x=0$. We can also write $-x$ for $(-1)x$ and x/α for $(1/\alpha)x$.

The Dot Product We shall have need for a notion of *dot product* of two vectors x and y. This is defined by

$$x\cdot y=x_1y_1+\cdots+x_ny_n, \tag{3}$$

where $x=(x_1,\ldots,x_n)$ and $y=(y_1,\ldots,y_n)$. Observe that the dot product (3) is a scalar. For that reason $x\cdot y$ is sometimes called the *scalar product* of x and y. It is also sometimes called the *inner product* of x and y. The dot product satisfies certain conditions that we summarize in Theorem 11.2.

Theorem 11.2: *If* $x, y, z\in\mathbb{R}^n$ *and* $\alpha\in\mathbb{R}$, *then*

$x\cdot y=y\cdot x$	*(commutative law)*
$x\cdot(y+z)=x\cdot y+x\cdot z$	*(distributive law)*
$(\alpha x)\cdot y=\alpha(x\cdot y).$	

We leave the proof as Exercise 11.1.2.

Observe that $0\cdot x=0$, but that $x\cdot y$ can equal 0 without either x or y being 0 [E.g., in $\mathbb{R}^2$, $(1,1)\cdot(1,-1)=1-1=0$]. Two nonzero vectors x and y for which $x\cdot y=0$ are said to be *orthogonal* vectors. Geometrically, two orthogonal vectors in $\mathbb{R}^n$ are perpendicular.

Exercises

11.1.1 Prove Theorem 11.1.

11.1.2 Prove Theorem 11.2.

11.1.3 Verify that $x\cdot x=0$ if and only if $x=0$.

11.1.4 Let $\mathcal{C}[a,b]$ be the set of continuous real functions on $[a,b]$. Verify that $\mathcal{C}[a,b]$ is a vector space under the usual notions of addition and scalar multiplication.

11.1.5 Let $\mathcal{D}[a,b]$ be the set of functions with the intermediate value property (IVP) on $[a,b]$. Is $\mathcal{D}$ a vector space under the usual notions of addition and scalar multiplication?
SEE NOTE 262

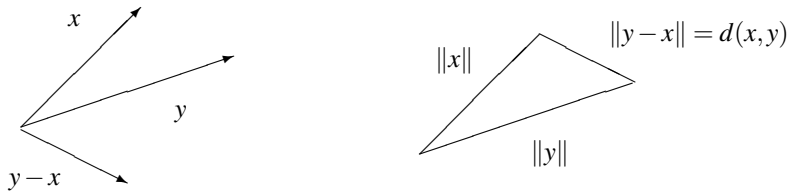


Figure 11.1. Vectors, distance, and norm in $\mathbb{R}^2$.

11.1.6 For f, g in $\mathcal{C}[0,2\pi]$, let $f \cdot g = \int_0^{2\pi} f(t)g(t)\,dt$.

(a) Show that "$\cdot$" satisfies the conditions of Theorem 11.2.

(b) Show that the set of functions

$$T = \{\sin nx, \cos mx : (n = 1,2,\dots\ m = 1,2,\dots\}$$

form an orthogonal set of functions in $\mathcal{C}$, that is, $f \cdot g = 0$ if $f, g \in T$, with $f \neq g$.

SEE NOTE 263

11.2 The Metric Structure of $\mathbb{R}^n$

The spaces $\mathbb{R}^n$ possess a natural notion of distance between two points. The notion of distance is fundamental to such notions as convergence of sequences, limits, continuity, and differentiation.

Our starting point is that of the norm $\|x\|$ of a vector in $\mathbb{R}^n$.

Definition 11.3: Let $x = (x_1, \dots, x_n) \in \mathbb{R}^n$. We define the *euclidean norm* by

$$\|x\| = \sqrt{\sum_{i=1}^{n} x_i^2}.$$

Note that $\|x\| \geq 0$ for all $x \in \mathbb{R}^n$.

In the familiar cases of $n = 1$, 2 or 3, $\|x\|$ is just the distance between x and the vector 0. [For example, in $\mathbb{R}^2$ with $z = (x,y)$ and $\|z\| = \sqrt{x^2 + y^2}$.]

This suggests the following definition of distance between the vectors x and y in $\mathbb{R}^n$ (Fig. 11.1).

Definition 11.4: For $x, y \in \mathbb{R}^n$, the *euclidean distance* between x and y is

$$d(x,y) = \|x - y\|.$$

In terms of coordinates, with $x = (x_1, \dots, x_n)$ and $y = (y_1, \dots, y_n)$

$$\|x - y\| = \sqrt{(x_1 - y_1)^2 + \cdots + (x_n - y_n)^2}.$$

Observe that we can express the norm via the dot product:

$$\|x\|^2 = x \cdot x \quad \text{or} \quad \|x\| = \sqrt{x \cdot x}.$$

Because of this, many norm computations are simplified by employing the dot product. An important relation between the dot product and norms is given by the Cauchy-Schwarz inequality.

Theorem 11.5 (Cauchy-Schwarz) *For $x, y \in \mathbb{R}^n$,*

$$|x \cdot y| \leq \|x\| \, \|y\|.$$

Proof. First observe that if $t \in \mathbb{R}$, then

$$p(t) = (x - ty) \cdot (x - ty) \geq 0. \tag{4}$$

Using Theorem 11.2, we can rewrite (4) as

$$p(t) = x \cdot x - 2(x \cdot y)t + (y \cdot y)t^2 \geq 0. \tag{5}$$

The function $p(t)$ is a quadratic in t. Since it is nonnegative for all t, it cannot have two distinct roots. Hence its discriminant cannot be positive, that is,

$$4(x \cdot y)^2 - 4(x \cdot x)(y \cdot y) \leq 0. \tag{6}$$

It now follows from (6) that

$$(x \cdot y)^2 \leq (x \cdot x)(y \cdot y)$$

and so

$$|x \cdot y| \leq \|x\| \, \|y\|$$

as was to be proved. ■

The Cauchy-Schwarz inequality allows us to obtain some important properties of the norm and of distance (or *metric* as it is often known).

Theorem 11.6 (Norm Properties) *Let $x, y \in \mathbb{R}^n$, and $\alpha \in \mathbb{R}$. Then*

(i) $\|x\| \geq 0$, *and* $\|x\| = 0$ *if and only if* $x = 0$.

(ii) $\|\alpha x\| = |\alpha| \, \|x\|$.

(iii) $\|x + y\| \leq \|x\| + \|y\|$ *(triangle inequality).*

Proof. The proofs of (i) and (ii) are immediate. To prove (iii), we calculate

$$\begin{aligned} \|x+y\|^2 &= (x+y) \cdot (x+y) = \|x\|^2 + 2x \cdot y + \|y\|^2 \\ &\leq \|x\|^2 + 2\|x\|\|y\| + \|y\|^2 = (\|x\|^2 + \|y\|^2)^2, \end{aligned}$$

the inequality following from the Cauchy-Schwarz inequality. ■

Theorem 11.7 (Metric Properties) *The distance function*

$$d : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$$

satisfies the following conditions for all $x, y, z \in \mathbb{R}^n$.

(i) $d(x, y) \geq 0$, *and* $d(x, y) = 0$ *if and only if* $x = y$.

(ii) $d(x,y) = d(y,x)$ *(symmetry).*

(iii) $d(x,z) \le d(x,y) + d(y,z)$ *(triangle inequality).*

Proof. Again, the proofs of (i) and (ii) are immediate. To verify (iii), let $u = x - y$, $v = y - z$. Then $u + v = x - z$. From (iii) of Theorem 11.6 we see that

$$\begin{aligned} \|u+v\| &\le \|u\| + \|v\|, \text{ or} \\ \|x-z\| &\le \|x-y\| + \|y-z\|, \end{aligned}$$

which is just the statement $d(x,z) \le d(x,y) + d(y,z)$. ■

In Exercises 11.2.6 and 11.2.7 we introduce two further norms on $\mathbb{R}^n$ and find some comparisons among them. Later, Exercises 11.3.8 and 11.4.1 show some ways in which these norms can be used interchangeably when dealing with concepts directly related to open and closed sets and convergence. Thus even though the euclidean norm encountered in this section is our main focus, other norms can be used at the same time to simplify and clarify the arguments.

Exercises

11.2.1 Establish the identity

$$\left(\sum_{i=1}^{n} x_i y_i\right)^2 = \left(\sum_{i=1}^{n} x_i^2\right)\left(\sum_{i=1}^{n} y_i^2\right) - \sum_{i<j} (x_i y_j - x_j y_i)^2$$

and use it to obtain another proof of the Cauchy-Schwarz inequality.

11.2.2 Verify the validity of parts (i) and (ii) in Theorems 11.6 and 11.7.

11.2.3 Prove that for $x, y \in \mathbb{R}^n$, $\|x-y\| \ge |\,\|x\| - \|y\|\,|$.

11.2.4 Let x, y be vectors in $\mathbb{R}^2$, and let θ be the angle between them. Use the law of cosines to verify that

$$x \cdot y = \|x\|\,\|y\| \cos\theta.$$

Thus, the dot product $x \cdot y$ equals the length of x times the (signed) length of the projection of y onto the vector x.

11.2.5 The result of Exercise 11.2.4 suggests a definition for the cosine of an angle between two nonzero vectors in $\mathbb{R}^n$:

$$\cos\theta = \frac{x \cdot y}{\|x\|\,\|y\|} \quad (0 \le \theta \le \pi).$$

The preceding definition of $\cos\theta$ is given via dot products and norms. Why does that imply $|\cos\theta| \le 1$?

11.2.6 The euclidean norm $\|x\| = \sqrt{x_1^2 + \cdots + x_n^2}$ is not the only norm that shall concern us. In the next chapter we shall find it computationally convenient to use the norm $\|x\|_1 = |x_1| + \cdots + |x_n|$.

(a) Prove that $\|x\|_1$ satisfies the conditions (i), (ii), (iii) of Theorem 11.6.

(b) If $d_1(x,y) = \|x-y\|_1$, show that $d_1(x,y)$ satisfies conditions (i), (ii), (iii) of Theorem 11.7.

(c) Show that for each $x \in \mathbb{R}^n$, $\|x\| \le \|x\|_1 \le \sqrt{n}\|x\|$.

11.2.7 There are many norms on $\mathbb{R}^n$ other than the euclidean norm and the norm $\|\cdot\|_1$ of Exercise 11.2.6. For "$\|\cdot\|$" to be a norm means that the three conditions of Theorem 11.6 are satisfied.

(a) Show that $\|x\|_\infty = \max(|x_1|, \ldots, |x_n|)$ is a norm on $\mathbb{R}^n$.

(b) Of the three norms $\|\cdot\|$, $\|\cdot\|_1$, and $\|\cdot\|_\infty$, for which is the Cauchy-Schwarz inequality valid?

(c) For which of these norms is it true that $|x \cdot x| = \|x\|^2$?

SEE NOTE 264

11.2.8 The identity

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2)$$

is known as the *parallelogram law.*

(a) Prove the identity is valid for all $x, y \in \mathbb{R}^n$.

(b) Interpret this identity as a statement about the sides and the diagonals of a parallelogram.

(c) Is the identity valid if we replace $\|\cdot\|$ by $\|\cdot\|_1$ or $\|\cdot\|_\infty$?

SEE NOTE 265

11.3 Elementary Topology of $\mathbb{R}^n$

By now you understand the fundamental role that the notion of distance plays in the study of $\mathbb{R}$. Indeed, the concept of open set is defined directly in terms of the distance notion. Various other important concepts, such as closed set, dense set, and accumulation point, depend directly or indirectly on the notion of distance. The same is true in $\mathbb{R}^n$. In this section we define some of these important concepts. We shall see that the definitions for $\mathbb{R}^n$ are the same as those for $\mathbb{R}$ if we replace the usual distance function on $\mathbb{R}$ by the euclidean distance function on $\mathbb{R}^n$.

Let $x_0 \in \mathbb{R}^n$, and let $r > 0$. The set

$$B(x_0, r) = \{x : \|x - x_0\| < r\}$$

is called the *open ball of radius r centered at* x_0. It is also called the *r-neighborhood* of x_0. It obviously consists of those points in $\mathbb{R}^n$ whose distance from x_0 is less than r. We define a number of terms, familiar in our study of $\mathbb{R}$, in terms of this simple concept.

1. A set E in $\mathbb{R}^n$ is *open* if to each $x_0 \in E$ there corresponds an $\varepsilon > 0$ such that $B(x_0, \varepsilon) \subset E$. An open set containing a point x_0 is also called a *neighborhood* of x_0.

2. A set E in $\mathbb{R}^n$ is *closed* if $\mathbb{R}^n \setminus E$ is open.

3. A point $x_0 \in E \subset \mathbb{R}^n$ is an *accumulation point*, or *limit point*, of E if every open ball centered at x_0 contains points of E other than x_0. We shall use the terms "accumulation point" and "limit point" interchangeably.

4. A point $x_0 \in \mathbb{R}^n$ is a *boundary point* of a set $E \subset \mathbb{R}^n$ provided that for each $\varepsilon > 0$, $B(x_0,\varepsilon) \cap E \neq \emptyset$ and $B(x_0,\varepsilon) \setminus E \neq \emptyset$.

5. A set $E \subset \mathbb{R}^n$ is *bounded* if there exists $M > 0$ such that
$$E \subset B(0,M).$$

We list in the next four examples some subsets of $\mathbb{R}^2$ and indicate which of the preceding properties they possess. We leave verification to you.

Example 11.8: The x-axis, $E_1 = \{(x,y) \in \mathbb{R}^2 : y = 0\}$. E_1 is not open; E_1 is closed; every point of E_1 is an accumulation point and a boundary point of E_1; E_1 is not bounded. ◀

Example 11.9: Let $E_2 = \{(x,y) : x^2 + y^2 \leq 1\}$. This set is not open; it is closed; every point of E_2 is an accumulation point of E_2; the boundary points of E_2 are those for which $x^2 + y^2 = 1$. E_2 is bounded. ◀

Example 11.10: The set
$$E_3 = \mathbb{Q} \times \mathbb{Q} = \{(x,y) : \ x \text{ and } y \text{ are rational}\}$$
is neither open nor closed; every point in $\mathbb{R}^2$ is an accumulation point of E_3 and a boundary point of E_3; E_3 is not bounded in $\mathbb{R}^2$. ◀

Example 11.11: The annular region
$$E_4 = \{(x,y) : 1 < x^2 + y^2 \leq 2\}$$
is neither open nor closed; every point in $\{(x,y) : 1 \leq x^2 + y^2 \leq 2\}$ is an accumulation point of E_4; the boundary points of E_4 are the points (x,y) such that $x^2 + y^2 = 1$ or $x^2 + y^2 = 2$; E_4 is bounded. ◀

Other terms we defined for $\mathbb{R}$ can be defined in the same way for $\mathbb{R}^n$. For example, the concepts of isolated point, dense set, and nowhere dense set can be defined in $\mathbb{R}^n$ in exactly the same way as they are in $\mathbb{R}$. Our development in the rest of this chapter and the next will deal primarily with open or closed sets in $\mathbb{R}^n$. We leave discussions of these and other concepts to the exercises. The notion of compactness will also be important to our development. We shall discuss compactness in Section 11.8.

A number of facts that relate the concepts in this section appear in the exercises. These are analogous to ones we already noted for $\mathbb{R}$.

Exercises

11.3.1 Show that a set $E \subset \mathbb{R}^n$ is closed if and only if it contains all of its limit points.

11.3.2 The *closure* $\overline{E}$ of a set E in $\mathbb{R}^n$ is the union of E and its accumulation points.

(a) Prove that E is closed if and only if $E = \overline{E}$.

(b) Prove that $\overline{B(x,\varepsilon)} = \{y : \|y - x\| \leq \varepsilon\}$.

11.3.3 List all subsets of $\mathbb{R}^n$ that are both open and closed.

11.3.4 Prove the following analogues of Theorems 4.17 and 4.18 in the setting of $\mathbb{R}^n$.

(a) The sets $\emptyset$ and $\mathbb{R}^n$ are both open and closed.

(b) Any intersection of a finite number of open sets is open.

(c) Any union of an arbitrary collection of open sets is open.

(d) The complement of an open set is closed.

(e) Any union of a finite number of closed sets is closed.

(f) Any intersection of an arbitrary collection of closed sets is closed.

(g) The complement of a closed set is open.

11.3.5 Let I be an open interval in $\mathbb{R}$, and let J be a closed interval in $\mathbb{R}$. We can view I and J as subsets of $\mathbb{R}^2$ by defining

$$I_1 = \{(x,0) : x \in I\} \text{ and } J_1 = \{(x,0) : x \in J\}.$$

Is I_1 open in $\mathbb{R}^2$? Is J_1 closed in $\mathbb{R}^2$?

11.3.6 ✂ Provide definitions for *isolated point*, *dense set*, and *nowhere dense set* in the setting of $\mathbb{R}^n$. The definitions should be analogous to those that apply in $\mathbb{R}$ so that the definitions in $\mathbb{R}$ are just special cases of those for $\mathbb{R}^n$.

(a) For each of the sets in Examples 11.8–11.11 indicate whether they have isolated points and whether they are dense or nowhere dense.

(b) Prove that E is dense if and only if $\overline{E} = \mathbb{R}^n$. (See Exercise 11.3.2.)

(c) Prove that E is nowhere dense if and only if every point of $\overline{E}$ is a boundary point of $\overline{E}$.

11.3.7 An open set D in $\mathbb{R}^n$ is called *connected* if to each pair of points x and y in D corresponds a "polygonal arc" in D, that is, a path consisting of a finite number of line segments joined end to end consecutively. Which of the following subsets of $\mathbb{R}^2$ are connected?

(a) The open unit ball centered at 0

(b) The set $D = \{(x,y) : xy > 0\}$

(c) The set $E = \{(x,y) : |y| > x^2\}$

SEE NOTE 266

11.3.8 Suppose we had defined $B_1(x_0,\varepsilon) = \{x : \|x - x_0\|_1 < \varepsilon\}$, replacing the norm $\|\cdot\|$ by $\|\cdot\|_1$.

(a) Sketch a picture of $B_1(0,1)$ in $\mathbb{R}^2$.

(b) Prove that a set $E \subset \mathbb{R}^n$ is open if and only if to each $x_0 \in E$ there corresponds $\varepsilon > 0$ such that $B_1(x_0,\varepsilon) \subset E$. (Thus $\mathbb{R}^n$ has exactly the same open sets when we use $\|\cdot\|_1$ as when we use $\|\cdot\|$ in the definition of open sets.)

(c) Are the closed sets exactly the same?

(d) Does the status of a point (as an accumulation point or boundary point or isolated point of a set) depend on whether we use $\|\cdot\|$ or $\|\cdot\|_1$ in our definition of open set?

(e) What about the status of a set as dense, nowhere dense, or bounded?

SEE NOTE 267

11.4 Sequences in $\mathbb{R}^n$

Much of what we have studied about sequences of real numbers carries over to the case of sequences in $\mathbb{R}^n$. Indeed, many of the proofs for sequences in $\mathbb{R}^n$ are virtually identical to the proofs of corresponding statements for sequences in $\mathbb{R}$. We shall leave most of these proofs to the exercises. When a proof requires a fresh idea, we shall provide the proof in detail.

Definition 11.12: A *sequence of points in* $\mathbb{R}^n$ is a function

$$f : \mathbb{N} \to \mathbb{R}^n.$$

As was the case for sequences in $\mathbb{R}$, we will frequently use notation such as $\{x_k\}$ to denote a sequence in $\mathbb{R}^n$. Note that x is boldface (denoting a vector) and k is not boldface since it indicates a natural number. If it is more convenient we can also rewrite the sequence as an n-tuple using double subscript notation:

$$x_k = (x_{k1}, \ldots, x_{kn}).$$

Definition 11.13: A sequence $\{x_k\}$ in $\mathbb{R}^n$ is *bounded* if there exists $M \in \mathbb{N}$ such that $\|x_k\| \leq M$ for all $k \in \mathbb{N}$.

We can define convergence of a sequence $\{x_k\}$ in $\mathbb{R}^n$ in the same way as we did for sequences of real numbers (Definition 2.6). Note that here the norm plays the same role that the absolute value did earlier.

Definition 11.14: Let $\{x_k\}$ be a sequence in $\mathbb{R}^n$. We say $\{x_k\}$ *converges* to a point x and write

$$\lim_{k\to\infty} x_k = x \text{ or}$$
$$x_k \to x \text{ as } k \to \infty$$

provided that for each $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $\|x_k - x\| < \varepsilon$ whenever $k \geq N$.

This definition is equivalent to the requirement that every open ball centered at x contains all but a finite number of terms of the sequence: For each $\varepsilon > 0$ there exists N such that $x_k \in B(x, \varepsilon)$ for all $k \geq N$. It is also equivalent to the requirements

$$\|x_k - x\| \to 0$$

or

$$d(x_k, x) \to 0$$

as $k \to \infty$, where d is the euclidean distance.

Coordinate-Wise Convergence We can also describe convergence in $\mathbb{R}^n$ in terms of coordinate-wise convergence. To see this, observe first that for $x = (x_1, \ldots, x_n)$ and $j = 1, \ldots, n$,

$$x_j^2 \leq \sum_{i=1}^{n} x_i^2 = \|x\|^2 \leq \left(\sum_{i=1}^{n} |x_i| \right)^2.$$

Thus

$$|x_j| \le \|x\| \le \sum_{i=1}^{n} |x_i|. \tag{7}$$

Now let

$$\{x_k\} = \{(x_{k1}, \ldots, x_{kn})\}$$

be a sequence in $\mathbb{R}^n$, and let $x = (x_1, \ldots, x_n)$ be a point in $\mathbb{R}^n$. By (7)

$$|x_{kj} - x_j| \le \|x_k - x\| \le \sum_{i=1}^{n} |x_{ki} - x_i|. \tag{8}$$

If $x_k \to x$ as $k \to \infty$, that is,

$$\|x_k - x\| \to 0 \quad \text{as } k \to \infty,$$

then we see from (8) that for each $j = 1, \ldots, n$, $|x_{kj} - x_j| \to 0$ as $k \to \infty$. Conversely, if for each $j = 1, \ldots, n$ we have $|x_{kj} - x_j| \to 0$ as $k \to \infty$, then

$$\sum_{i=1}^{n} |x_{ki} - x_i| \to 0$$

as $k \to \infty$, so we see once again from (8) that $\|x_k - x\| \to 0$ as $k \to \infty$. We summarize this discussion as a theorem.

Theorem 11.15: *Let*

$$\{x_k\} = \{(x_{k1}, \ldots, x_{kn})\}$$

be a sequence in $\mathbb{R}^n$ *and let* $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$. *Then*

$$\lim_{k\to\infty} x_k = x$$

if and only if for each $j = 1, \ldots, n$, $\lim_{k\to\infty} x_{kj} = x_j$.

Algebraic Properties Our next result shows that limits combine as expected with respect to addition, multiplication by a scalar, and the dot product operation. We leave the proof as Exercise 11.4.2.

Theorem 11.16: *Let* $\{x_k\}$ *and* $\{y_k\}$ *be sequences in* $\mathbb{R}^n$, *and let* $\alpha \in \mathbb{R}$. *If* $\lim_{k\to\infty} x_k = x$ *and* $\lim_{k\to\infty} y_k = y$, *then*

(i) $\lim_{k\to\infty} (\alpha x_k) = \alpha x$.

(ii) $\lim_{k\to\infty} (x_k + y_k) = x + y$.

(iii) $\lim_{k\to\infty} (x_k \cdot y_k) = x \cdot y$.

Limit Points As was true in $\mathbb{R}$, we can characterize limit points in $\mathbb{R}^n$ in the language of sequences.

Theorem 11.17: *Let* $E \subset \mathbb{R}^n$ *and let* $x \in \mathbb{R}^n$. *Then* x *is a limit point of* E *if and only if there exists a sequence* $\{x_k\}$ *of distinct points of* E *such that* $\lim_{k\to\infty} x_k = x$.

Proof. Suppose x is a limit point of E. Then for each $\varepsilon > 0$ there exists $x_k \neq x$ in E such that $\|x_k - x\| < \varepsilon$. Choose $x_1 \in E$ such that $0 < \|x_1 - x\| < 1$. Inductively, having chosen x_k, choose x_{k+1} in E such that

$$0 < \|x_{k+1} - x\| < \tfrac{1}{2}\|x_k - x\|.$$

Then $\lim_{k\to\infty} x_k = x$, $x_k \neq x$, and if $k \neq j$, $x_k \neq x_j$. The converse is obvious. ■

Theorem 11.18: *A set $E \subset \mathbb{R}^n$ is closed if and only if it contains all its limit points.*

The proof is left as Exercise 11.4.3.

Bolzano-Weierstrass Theorem The Bolzano-Weierstrass Theorem in one dimension asserts that bounded sequences of real numbers must have convergent subsequences. This theorem carries over to $\mathbb{R}^n$ and will play an important role in our study of continuous functions on $\mathbb{R}^n$. The proof involves using the one-dimensional version n times. While the idea of the proof is straightforward, the notation is messy, involving subsequences of subsequences of subsequences, n times. We shall adopt more condensed notation in order to avoid multiple levels of subscripts. It is worth studying the proof to see how messy details can be simplified by an appropriate notation.

Theorem 11.19 (Bolzano-Weierstrass) *Every bounded sequence $\{x_k\}$ in $\mathbb{R}^n$ contains a convergent subsequence.*

Proof. Let $\{x_k\}$ be a bounded sequence in $\mathbb{R}^n$, say $\|x_k\| \leq M$ for all $k \in \mathbb{N}$. For each k, let $x_k = (x_{k1}, \ldots, x_{kn})$. Then for each $k \in \mathbb{N}$ and $j = 1, \ldots, n$, $|x_{kj}| \leq M$. Thus, for all j, the sequence of real numbers $\{x_{kj}\}$ is bounded.

Let $j = 1$. By the Bolzano-Weierstrass theorem applied to the bounded sequence $\{x_{k1}\}$ $(k = 1, 2, 3, \ldots)$, there exists a sequence of integers

$$1 \leq k(1,1) < k(1,2) < \ldots$$

and a number x_1 such that

$$x_{k(1,i),1} \to x_1 \text{ as } i \to \infty.$$

Observe that the sequence $\{x_{k(1,i),1}\}$ is just a subsequence of the sequence $\{x_{k1}\}$.

Next let $j = 2$. The sequence $\{x_{k(1,i),2}\}$ is also bounded by M, so again by the Bolzano-Weierstrass theorem, there is a subsequence $\{k(2,i)\}$ of $\{k(1,i)\}$ such that $\{x_{k(2,i),2}\}$ converges. Let

$$x_2 = \lim_{i\to\infty} x_{k(2,i),2}.$$

Now, the sequence $\{x_{k(1,i),1}\}$ converges to x_1, so the same is true of any subsequence of this sequence. In particular $\{x_{k(2,i),1}\} \to x_1$ as $i \to \infty$.

We continue this process. Having obtained a sequence $\{x_{k(j,i),j}\}$ with

$$\{x_{k(m,i),m}\} \to x_m \text{ for all } m \leq j,$$

we use the Bolzano-Weierstrass theorem yet again to obtain a further subsequence $\{x_{k(j+1,i),j+1}\}$ that converges to a point $x_{j+1} \in \mathbb{R}$. The process stops when $j + 1 = n$. Letting $k_i = k(n,i)$ and $x = (x_1, \ldots, x_n)$, we have $\lim_{i\to\infty} x_{k_i} = x$ by Theorem 11.15. ■

Exercises

11.4.1 For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ define the norms

$$\|x\| = \sqrt{x_1^2 + \cdots + x_n^2},$$
$$\|x\|_1 = |x_1| + \cdots + |x_m|,$$
$$\|x\|_\infty = \max\{|x_i| : i = 1, \ldots, n\}.$$

Prove that the following are equivalent for sequences in $\mathbb{R}^n$.

(a) $\lim_{k\to\infty} x_k = x$

(b) $\|x_k - x\|_1 \to 0$ as $k \to \infty$

(c) $\|x_k - x\|_\infty \to 0$ as $k \to \infty$

Thus convergence does not depend on which of the norms $\|\cdot\|$, $\|\cdot\|_1$, $\|\cdot\|_\infty$ we use.

11.4.2 Prove Theorem 11.16.

11.4.3 Prove Theorem 11.18.

11.4.4 Prove that every convergent sequence in $\mathbb{R}^n$ is bounded.

11.4.5 Prove that a set $E \subset \mathbb{R}^n$ is closed and bounded if and only if every sequence $\{x_k\}$ of points of E has a subsequence converging to a point in E.

11.4.6 State and prove the analogue in $\mathbb{R}^n$ to Cauchy's criterion for the convergence of real sequences (Theorem 2.41).

11.4.7 Which of the following sequences $\{x_k\}$ in $\mathbb{R}^3$ or $\mathbb{R}^2$ are convergent?

(a) $x_k = (\cos\frac{\pi}{k}, \sin\frac{\pi}{k}, \frac{k}{1+k})$

(b) $x_k = (\cos\pi k, \sin\pi k, \frac{k}{k+1})$

(c) $x_k = (e^{-k}, \frac{1}{k!}, k)$

(d) $x_k = (\frac{\ln k}{k}, \frac{k^2}{k^2+1}, (-1)^k)$

(e) $x_k = (\frac{\sin k}{k}, k\sin\frac{1}{k})$

(f) $x_k = (\sqrt{k+1} - \sqrt{k}, 7)$

11.4.8 Which of the sequences in Exercise 11.4.7 have convergent subsequences?

11.4.9 Prove that if $\{x_k\} \to x$ as $k \to \infty$, then $\|x_k\| \to \|x\|$ as $k \to \infty$. Does this conclusion remain valid if the euclidean norm is replaced by the norms $\|\cdot\|_1$ or $\|\cdot\|_\infty$ of Exercise 11.4.1?

11.4.10 Prove the version of the Bolzano-Weierstrass theorem that applies to sets: Every infinite bounded subset of $\mathbb{R}^n$ has a point of accumulation in $\mathbb{R}^n$.

11.5 Functions and Mappings

When we dealt with functions in the preceding chapters, we were concerned primarily with functions whose domains were subsets of $\mathbb{R}$ and whose ranges were in $\mathbb{R}$; these functions have traditionally been known as "real-valued functions of a real variable." The domains were usually intervals. In this chapter and the next we are concerned with

functions whose domains and ranges are in the spaces $\mathbb{R}^n$. Usually, the domain will be an open subset of $\mathbb{R}^n$. As was the case with real functions of a real variable, our study focuses on questions concerned with limits, continuity, and differentiability. In the present section we deal only with some definitions, some examples, and a bit about notation.

11.5.1 Functions from $\mathbb{R}^n \to \mathbb{R}$

Let us begin with functions $f : E \to \mathbb{R}$, where E is a subset of $\mathbb{R}^n$. Such functions are sometimes called "real functions of several variables." We present some examples.

There is a special and traditional notational feature for the case $n = 2$ (i.e., for functions $f : \mathbb{R}^2 \to \mathbb{R}$). The two variables needed to represent a point $(x_1, x_2) \in \mathbb{R}^2$ are written as (x, y). In all discussions the x refers to the first variable and the y to the second. Similarly for functions $f : \mathbb{R}^3 \to \mathbb{R}$ we often use (x, y, z) rather than (x_1, x_2, x_3) to represent points in $\mathbb{R}^3$. We will usually make use of this convention, especially in discussions of derivatives in the next chapter. For $n \geq 4$ the subscript notation is used.

Example 11.20: Let $f(x, y) = x^2 + xy + y^2 + 5$. A natural domain for f is all of $\mathbb{R}^2$, so $f : \mathbb{R}^2 \to \mathbb{R}$ is a function of two variables. The function f is an example of a polynomial in two variables. The general polynomial in two variables can be written as

$$f(x, y) = \sum_{i,j} a_{ij} x^i y^j \tag{9}$$

where the coefficients a_{ij} are real numbers, the numbers i and j are nonnegative integers, and the sum has finitely many terms. Figure 11.2 shows the graph of this polynomial f in $\mathbb{R}^3$; you should be familiar with such representations from calculus classes. For much of our study we shall not be able to rely on any meaningful pictures since we study functions of many variables, and pictures outside of $\mathbb{R}^2$ and $\mathbb{R}^3$ are not possible. Even so, we rely on these special cases to help develop our intuition as to what is going on in more general situations. ◀

Example 11.21: Let $g(x_1, x_2, x_3) = \sqrt{1 - x_1^2 - x_2^2 - x_3^2}$. Here a natural domain for g is the set

$$E = \{(x_1, x_2, x_3) : x_1^2 + x_2^2 + x_3^2 \leq 1\}.$$

You will recognize E as the closed unit ball in the space $\mathbb{R}^3$. ◀

Example 11.22: Let $h(x, y) = \ln xy$. A natural domain for h is the set $E = \{(x, y) : xy > 0\}$. Thus E consists of the union of the first and third open quadrants of the xy plane. ◀

Example 11.23: To each triangle T contained in $\mathbb{R}^2$ let $A(T)$ denote the area of the triangle. We can view A as a function whose domain is the set of triangles in $\mathbb{R}^2$. But we can also express a formula for A as a function of six variables. If (a, b), (c, d), (e, f) are the vertices of T, then we can express the area as a function $g : \mathbb{R}^6 \to \mathbb{R}$ where $g(a, b, c, d, e, f) = A(T)$. (Exercise 11.5.2 asks for an analytic formula for the area.) ◀

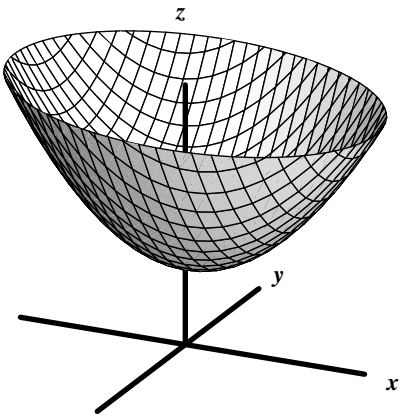


Figure 11.2. Graph of the polynomial $f(x,y) = x^2 + xy + y^2 + 5$.

Sometimes it is more convenient to represent a function without referring directly to the coordinates.

Example 11.24: Let $a \in \mathbb{R}^n$ and let $f : \mathbb{R}^n \to \mathbb{R}$ be defined by $f(x) = a \cdot x$. If $a = (a_1, \ldots, a_n)$ and $x = (x_1, \ldots, x_n)$, then

$$f(x) = f(x_1, \ldots, x_n) = a_1 x_1 + \cdots + a_n x_n.$$

Functions of this type are *linear* functions. This means they satisfy the condition

$$f(\alpha x + \beta y) = \alpha f(x) + \beta f(y) \text{ for all } \alpha, \beta \in \mathbb{R} \text{ and all } x, y \in \mathbb{R}^n.$$

In fact *every* linear function $f : \mathbb{R}^n \to \mathbb{R}$ can be represented in the form $f(x) = a \cdot x$ for some $a \in \mathbb{R}^n$ and all $x \in \mathbb{R}^n$. We leave verification as Exercise 11.5.3. ◀

Example 11.25: For each $x \in \mathbb{R}^n$, let $f(x) = \|x\|$. Then $f : \mathbb{R}^n \to \mathbb{R}$. In terms of coordinates

$$f(x) = f(x_1, \ldots, x_n) = \sqrt{x_1^2 + \cdots + x_n^2}.$$

◀

Exercises

11.5.1 Give a definition for a polynomial of n-variables. Write the definition in a form analogous to equation (9).

11.5.2 Refer to Example 11.23. Find a formula for $A(T)$ as a function of six variables.

11.5.3 Refer to Example 11.24. Prove that the function $f : \mathbb{R}^n \to \mathbb{R}$ is linear if and only if there exists a unique $a \in \mathbb{R}^n$ such that $f(x) = a \cdot x$ for all $x \in \mathbb{R}^n$.

11.5.2 Functions from $\mathbb{R}^n \to \mathbb{R}^m$

When dealing with functions with domain in $\mathbb{R}^n$ and range in $\mathbb{R}^m$, we often use the term *mapping* or *transformation* or *operator* in place of "function." The term *vector-valued function* is also in common use.

We have been denoting vectors or points in $\mathbb{R}^m$ $(m \geq 2)$ with **bold** letters. We shall also denote functions with range in $\mathbb{R}^m$ $(m \geq 2)$ with **bold** letters. Suppose $E \subset \mathbb{R}^n$ and $f : E \to \mathbb{R}^m$. Then for all $x \in E$, $f(x) \in \mathbb{R}^m$. If we express x and $f(x)$ in terms of coordinates, say $x = (x_1, \ldots, x_n)$ and $f(x) = (y_1, \ldots, y_m)$, then

$$f(x_1, \ldots, x_n) = (y_1, \ldots, y_m).$$

The numbers $y_1, \ldots, y_m$ depend on $x = (x_1, \ldots, x_n)$, of course. Thus there exist functions $f^1, \ldots, f^m : E \to \mathbb{R}$ called the *coordinate functions* such that

$$\begin{aligned} y_1 &= f^1(x) = f^1(x_1, \ldots, x_n) \\ y_2 &= f^2(x) = f^2(x_1, \ldots, x_n) \\ &\vdots \\ y_m &= f^m(x) = f^m(x_1, \ldots, x_n) \end{aligned}$$

We use superscripts for the functions f^j instead of subscripts in order to avoid confusion with the subscript notation we shall use in Chapter 12 for partial derivatives.

Example 11.26: Define $f : \mathbb{R}^2 \to \mathbb{R}^2$ by

$$f(r, \theta) = (r\cos\theta, r\sin\theta), \quad r, \theta \in \mathbb{R}.$$

Then $f^1(r, \theta) = r\cos\theta$, $f^2(r, \theta) = r\sin\theta$. For $r \geq 0$, this mapping f is familiar in connection with transforming polar coordinates to rectangular coordinates. Here we usually write x for f^1 and y for f^2, so we get

$$x = r\cos\theta \ , \ y = r\sin\theta.$$

◀

Example 11.27: (Linear functions) Suppose $A : \mathbb{R}^n \to \mathbb{R}^m$ is given by the equations

$$\begin{aligned} y_1 &= a_{11}x_1 + \cdots + a_{1n}x_n \\ y_2 &= a_{21}x_1 + \cdots + a_{2n}x_n \\ &\vdots \\ y_m &= a_{m1}x_1 + \cdots + a_{mn}x_n \end{aligned}$$

where $a_{ij} \in \mathbb{R}$, $i = 1, \ldots, m$, $j = 1, \ldots, n$ and $x = (x_1, \ldots, x_n)$. The transformation A can be represented by a matrix $A = (a_{ij})$, the m by n matrix of coefficients. We then have

$$A(x) = Ax,$$

the product being matrix multiplication:

$$\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \\ a_{m1} & \dots & a_{mn} \end{pmatrix} \begin{pmatrix} x_1 \\ \vdots \\ \vdots \\ x_n \end{pmatrix}$$

It is easy to check that A is linear, that is, that

$$A(\alpha x_1 + \beta x_2) = \alpha A(x_1) + \beta A(x_2)$$

for all $\alpha, \beta \in \mathbb{R}$ and all $x_1, x_2 \in \mathbb{R}^n$. (See Exercise 11.5.4.) ◀

Observe that in Example 11.27 we used capital letters to denote linear transformations from $\mathbb{R}^n \to \mathbb{R}^m$. This is a common practice that is also frequently used when dealing with linear transformations on other vector spaces (see Chapter 13). The use of capital letters for transformations from $\mathbb{R}^n$ to $\mathbb{R}^m$ is also sometimes preferred even when the functions are not linear. We shall use whatever notation is convenient for our purposes. One more word about notation. It is sometimes convenient, particularly when dealing with linear transformations, to use notation such as Ax in place of $A(x)$. We shall use whichever of these notations is convenient at the time.

Example 11.28: Let $T(x,y) = (u,v)$ where $u = e^x \cos y$, $v = e^x \sin y$. Thus $u = T^1(x,y) = e^x \cos y$ and $v = T^2(x,y) = e^x \sin y$. We can study some geometric properties of the mapping T.

Observe that $u^2 + v^2 = e^{2x}$, and $v \cos y = u \sin y$. For fixed x_0, (u,v) is a point on the circle having center $(0,0)$ and radius e^{x_0}. As y increases over an interval of length 2π, (u,v) traverses that circle one time. Thus T maps a line $x = x_0$ infinitely often around that circle. Similarly, T maps a line $y = y_0$ onto a ray having slope $\tan y_0$ and emanating from the origin.

We leave verification of this and some other mapping properties of T to Exercise 11.5.5. Note that this function offers us some problems should we wish a graphical representation. The graph would be a subset of $\mathbb{R}^4$ and so beyond our abilities to represent. Figure 11.3 can be used to study this function; you should be familiar with such representations from calculus classes. This picture carries the information discussed previously about lines mapping into circles. ◀

In Chapter 12 we shall be concerned with differentiability of functions of several variables and of mappings. Central to our work will be the use of linear transformations to approximate such functions or mappings.

Exercises

11.5.4 Refer to Example 11.27. Prove that A is linear.

11.5.5 Refer to Example 11.28. Verify that T has the mapping properties stated in that example. Show also that if S is the strip in the xy plane between the lines $y = 0$ and $y = 2\pi$, then $T(S) = \mathbb{R}^2 \setminus (0,0)$.

11.5.6 Let $H : \mathbb{R}^2 \to \mathbb{R}^2$ be defined by $H(x,y) = (x^2 - y^2, 2xy)$. Write $u = x^2 - y^2$ and $v = 2xy$, so that $H(x,y) = (u,v)$.

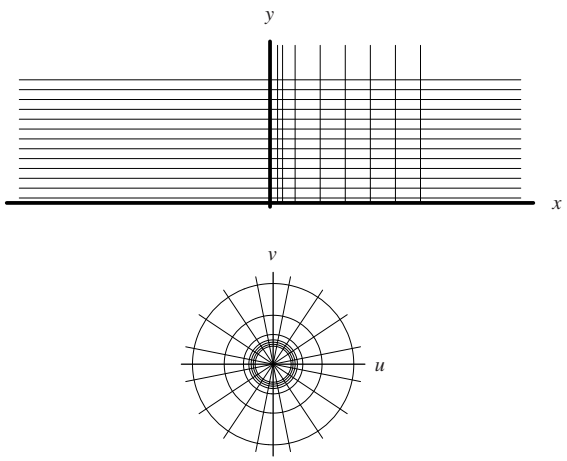


Figure 11.3. A representation of the function $T(x,y) = (u,v)$, where $u = e^x \cos y$, $v = e^x \sin y$.

(a) Describe the sets in the xy-plane that H maps onto horizontal lines in the uv-plane. Do the same for vertical lines.

(b) Determine a set $S \subset \mathbb{R}^2$ such that $H(S) = [1,2] \times [4,5]$.

11.6 Limits of Functions from $\mathbb{R}^n \to \mathbb{R}^m$

The concept of limit for a function $f : E \to \mathbb{R}^m$, where $E \subset \mathbb{R}^n$ is entirely analogous to the concept of limit for a real function $f : E \to \mathbb{R}$, where $E \subset \mathbb{R}$. We wish to capture the idea that $f(x)$ is near $y_0 \in \mathbb{R}^m$ when x is near x_0 in E. Our definition and some basic results and their proofs are essentially identical to their counterparts for real functions.

11.6.1 Definition

We begin with the ε-δ definition of the limit for functions from $\mathbb{R}^n$ to $\mathbb{R}^m$. The equivalence with a sequence version is proved in Lemma 11.30 which follows.

Definition 11.29: Let $f : E \to \mathbb{R}^m$, where E is a subset of $\mathbb{R}^n$ and let x_0 be a limit point of E. We say

$$\lim_{x \to x_0} f(x) = y_0$$

if for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|f(x) - y_0\| < \varepsilon$ whenever $0 < \|x - x_0\| < \delta$ and $x \in E$.

Observe that the existence of the limit, and its value, does not depend on whether f is or is not defined at x_0 or on its value at x_0. In loose geometric language, $\lim_{x \to x_0} f(x) = y_0$ means that to every neighborhood V of y_0, no matter how small, corresponds a sufficiently small neighborhood U of x_0 such that f maps $(U \cap E) \setminus \{x_0\}$ into V. (This definition and this discussion should be compared with the treatment of limits for functions of a single real variable given in Chapter 5.)

Our lemma relates this notion of limit with the sequential one (which is the analogue of the material in Section 5.1.2 for functions of one variable).

Lemma 11.30: *Let $E \subset \mathbb{R}^n$ and let x_0 be an accumulation point of E. Suppose $f : E \to \mathbb{R}^m$. Then $\lim_{x\to x_0} f(x) = y_0$ if and only if for every sequence $\{x_k\} \subset E$ such that $x_k \to x_0$, $x_k \neq x_0$, we have $f(x_k) \to y_0$.*

Proof. Suppose $\lim_{x\to x_0} f(x) = y_0$. Let $x_k \to x_0$ $(x_k \neq x_0)$, $x_k \in E$ and let $\varepsilon > 0$. Then there exists $\delta > 0$ such that if $x \in E$ and $0 < \|x - x_0\| < \delta$,

$$\|f(x) - y_0\| < \varepsilon.$$

Since $x_k \to x_0$, there exists $N \in \mathbb{N}$ such that $\|x_0 - x_k\| < \delta$ if $k \geq N$. Thus, for $k \geq N$,

$$\|f(x_k) - y_0\| < \varepsilon,$$

and $f(x_k) \to y_0$.

To prove the converse, suppose y_0 is *not* the limit of $f(x)$ as $x \to x_0$. Then there exists $\varepsilon > 0$ such that for each $k \in \mathbb{N}$ there exists $x_k \in E$ such that

$$0 < \|x_k - x_0\| < \frac{1}{k}$$

and

$$\|f(x_k) - y_0\| \geq \varepsilon.$$

The sequence $\{x_k\}$ converges to x_0 but the sequence $f(x_k)$ does not converge to y_0. ■

Example 11.31: Let

$$f(x,y) = xy\left(\frac{x^2 - y^2}{x^2 + y^2}\right),\ f(0,0) = 7.$$

Then $f : \mathbb{R}^2 \to \mathbb{R}$. We show that $\lim\limits_{(x,y)\to(0,0)} f(x,y) = 0$. Let $\varepsilon > 0$. We have

$$|f(x,y) - 0| \leq |xy|\frac{|x^2 - y^2|}{|x^2 + y^2|} \leq |xy|$$

when $(x,y) \neq (0,0)$. Now $|xy| \leq x^2 + y^2$ for all $(x,y) \in \mathbb{R}^2$, so $|f(x,y)| < \varepsilon$ whenever $\sqrt{x^2 + y^2} < \delta = \sqrt{\varepsilon}$, $(x,y) \neq (0,0)$. Thus

$$\lim_{(x,y)\to(0,0)} f(x,y) = 0.$$

Note that the fact that $f(0,0) = 7$ does not affect the limit. ◄

When a limit exists, it must be unique (Exercise 11.6.2). We can sometime use this fact to show that a limit does not exist.

Example 11.32: Let

$$f(x,y) = \frac{2xy}{x^2 + y^2},\ f(0,0) = 0.$$

Here $\lim_{y\to 0} f(0,y) = \lim_{x\to 0} f(x,0) = 0$, but $\lim\limits_{(x,y)\to(0,0)} f(x,y)$ does not exist. To see this, observe that

$$\lim_{(t,t)\to(0,0)} f(t,t) = \lim_{t\to 0} \frac{2t^2}{2t^2} = 1.$$

Thus, approaching $(0,0)$ along the coordinate axes requires the limit to be 0, but approaching $(0,0)$ along the line $y = x$ requires the limit to be 1. As a result, for $0 < \varepsilon < 1$, there is no $y_0 \in \mathbb{R}$ and $\delta > 0$ such that $|f(x,y) - y_0| < \varepsilon$ whenever $0 < \sqrt{x^2+y^2} < \delta$. ◀

Example 11.33: Let $f(r,\theta) = (r\cos\theta, r\sin\theta)$. Thus $f : \mathbb{R}^2 \to \mathbb{R}^2$. We show that

$$\lim_{(r,\theta)\to(0,0)} f(r,\theta) = (0,0).$$

Here

$$\|f(r,\theta) - (0,0)\| = \sqrt{r^2\cos^2\theta + r^2\sin^2\theta} = |r|.$$

Thus $\|f(r,\theta) - (0,0)\| < \varepsilon$ whenever $|r| < \varepsilon$, independent of θ. In particular if

$$0 < \|(r,\theta)\| = \sqrt{r^2+\theta^2} < \delta = \varepsilon,$$

then $|r| < \varepsilon$, so

$$\lim_{(r,\theta)\to(0,0)} f(r,\theta) = (0,0).$$

◀

Exercises

11.6.1 Establish Examples 11.31 and 11.32 by using Lemma 11.30 rather than an ε-δ argument.

11.6.2 **(Uniqueness of Limits)** Let $E \subset \mathbb{R}^n$ and let x_0 be an accumulation point of E. Suppose $f : E \to \mathbb{R}^m$. If

$$\lim_{x\to x_0} f(x) = y_0 \text{ and } \lim_{x\to x_0} f(x) = z_0,$$

then $y_0 = z_0$.

11.6.3 Let $E \subset \mathbb{R}^n$ and let x_0 be an accumulation point of E. Prove that if

$$\lim_{x\to x_0} f(x)$$

exists, then there exist $\varepsilon > 0$ and $M > 0$ such that $\|f(x)\| < M$ for all $x \in B(x_0,\varepsilon)$ at which f is defined.

11.6.2 Coordinate-Wise Convergence

We have seen earlier (Exercises 11.2.6, 11.2.7, 11.3.8, and 11.4.1) that there are further norms on $\mathbb{R}^n$ that can be used to study convergence. Let us briefly discuss the use of the norm $\|\cdot\|_\infty$ in connection with the limit concept. This will give us some insight into the nature of the limit concept and, in particular, allows us to show that convergence can be interpreted coordinate-wise.

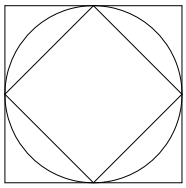


Figure 11.4. Three unit balls centered at the origin for the three norms $\|\cdot\|_1$, $\|\cdot\|$, and $\|\cdot\|_\infty$ (from innermost to outermost).

We start by recalling the definition of the infinity norm and comparing it to the euclidean norm. For $x = (x_1, \ldots, x_n) \in \mathbb{R}^n$ we defined

$$\|x\|_\infty = \max\{|x_1|, |x_2|, \ldots, |x_n|\}.$$

For simplicity, consider first the case of $\mathbb{R}^2$, using the notation (x,y) for members of $\mathbb{R}^2$. Then

$$\|(x,y)\| = \sqrt{x^2+y^2} \text{ while } \|(x,y)\|_\infty = \max\{|x|, |y|)\}.$$

The open euclidean ball $B((0,0),r)$ is an open disk centered at $(0,0)$ with radius r. The open ball $B_\infty((0,0),r)$ is an open square with center at $(0,0)$ having sides parallel to the coordinate axes and side length $2r$. Note that

$$B((0,0),r) \subset B_\infty((0,0),r),$$

and there exists $s > 0$ such that

$$B_\infty((0,0),s) \subset B((0,0),r).$$

Figure 11.4 shows the three unit balls centered at the origin for the three norms $\|\cdot\|_1$, $\|\cdot\|$, and $\|\cdot\|_\infty$. A similar comment applies to balls centered at other points of $\mathbb{R}^2$.

For an arbitrary $n \in \mathbb{N}$, something similar is true. For every $r > 0$ and $x_0 \in \mathbb{R}^n$, we have

$$\{x \in \mathbb{R}^n : \|x - x_0\| < r\} \subset \{x \in \mathbb{R}^n : \|x - x_0\|_\infty < r\} \tag{10}$$

and there exists $s > 0$ such that

$$\{x \in \mathbb{R}^n : \|x - x_0\|_\infty < s\} \subset \{x \in \mathbb{R}^n : \|x - x_0\| < r\}. \tag{11}$$

From these inclusions we can see that we could substitute $\|\cdot\|_\infty$ for $\|\cdot\|$ in Definition 11.14 and Definition 11.29 without changing any conclusions. That is, the definition of the sequence limit is equivalent to the following version with a different norm.

Lemma 11.34: *The limit $x_k \to x$ is valid in the sense of the norm $\|\cdot\|$ if and only if for all $\varepsilon > 0$, there exists an integer N such that*

$$\|x_k - x\|_\infty < \varepsilon$$

whenever $k \geq N$.

Proof. Suppose $x_k \to x$ and $\varepsilon > 0$. Then there exists $N \in \mathbb{N}$ such that $\|x_k - x\| < \varepsilon$ for all $k \geq N$. From (10) we see $\|x_k - x\|_\infty < \varepsilon$ for all $k \geq N$. Thus $x_k \to x$ relative to $\|\cdot\|_\infty$.

Conversely, suppose $x_k \to x$ relative to $\|\cdot\|_\infty$. Let $\varepsilon > 0$. Using (11), choose $\varepsilon' < \varepsilon$ such that if $\|z - x\|_\infty < \varepsilon'$, then $\|z - x\| < \varepsilon$. Now, there exists $N \in \mathbb{N}$ such that if $k \geq N$,

then $\|x_k - x\|_\infty < \varepsilon'$. Thus for $k \geq N$, $\|x_k - x\| < \varepsilon$. This shows that $x_k \to x$ relative to $\|\cdot\|$. ■

Similarly, the function limit can also be written using the other norm.

Lemma 11.35: *Let $E \subset \mathbb{R}^n$, $f : E \to \mathbb{R}^m$. Let $x_0 = (x_1, \ldots, x_n)$ be a limit point of E and let $y_0 = (y_1, \ldots, y_m) \in \mathbb{R}^m$. Then*

$$\lim_{x \to x_0} f(x) = y_0$$

if and only if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\|f(x) - y_0\|_\infty < \varepsilon$$

whenever $0 < \|x - x_0\|_\infty < \delta$ and $x \in E$.

We leave the proof as Exercise 11.6.5.

Coordinate-Wise Convergence As an application of the preceding remarks we show that convergence of maps from $\mathbb{R}^n$ to $\mathbb{R}^m$ reduces to coordinate-wise convergence.

Theorem 11.36: *Let $E \subset \mathbb{R}^n$, $f : E \to \mathbb{R}^m$, $f = (f^1, \ldots, f^m)$. Let $x_0 = (x_1, \ldots, x_n)$ be a limit point of E. Let $y_0 = (y_1, \ldots, y_m) \in \mathbb{R}^m$. Then $\lim_{x \to x_0} f(x) = y_0$ if and only if*

$$\lim_{x \to x_0} f^j(x) = y_j \text{ for all } j = 1, \ldots, m.$$

Proof. The proof follows immediately from Lemma 11.35 and the observation that for any $p \in \mathbb{N}$ and any vector $z = (z_1, \ldots, z_p) \in \mathbb{R}^p$, $|z_i| \leq \|z\|$ for all $i = 1, \ldots, p$, so $\|z\|_\infty \to 0$ is therefore equivalent to

$$\lim_{z \to z_0} z_i = 0 \text{ for all } i = 1, \ldots, p.$$

Thus, for example,

$$\lim_{x \to x_0} \|f(x) - y_0\|_\infty = 0$$

is equivalent to

$$\lim_{x \to x_0} |f^j(x) - y_j| = 0 \text{ for all } j = 1, \ldots, m.$$

■

Exercises 11.2.6, 11.2.7, 11.3.8, and 11.4.1 are all concerned with comparing several norms for $\mathbb{R}^n$. In essence, these exercises suggested that the norms $\|\cdot\|$, $\|\cdot\|_1$, and $\|\cdot\|_\infty$ could be used interchangeably when dealing with concepts directly related to open sets and to convergence.

Exercises

11.6.4 Verify the correctness of (10) and (11).

11.6.5 Prove Lemma 11.35.

SEE NOTE 268

11.6.3 Algebraic Properties

An immediate consequence of Theorem 11.36 is that some of the algebraic properties of limits we obtained in Section 5.2.3 for real functions of one real variable are also valid in our present setting.

Theorem 11.37: *Suppose that* $E \subset \mathbb{R}^n$,

$$f : E \to \mathbb{R}^m,\ g : E \to \mathbb{R}^m,$$

$\alpha \in \mathbb{R}$, *and* x_0 *is an accumulation point of* E. *If* $\lim_{x\to x_0} f(x) = y_0$ *and* $\lim_{x\to x_0} g(x) = z_0$, *then*

1. $\lim_{x\to x_0}(f(x)+g(x)) = y_0 + z_0$ *and*
2. $\lim_{x\to x_0} f(\alpha x) = \alpha y_0$.

We leave the proof to the exercises. Regarding products, the expected product rule for dot products is valid.

Theorem 11.38: *Let* $E \subset \mathbb{R}^n$, *let* x_0 *be an accumulation point of* E *and let* $f, g : E \to \mathbb{R}^m$. *If*

$$\lim_{x\to x_0} f(x) = y_0 \text{ and } \lim_{x\to x_0} g(x) = z_0,$$

then

$$\lim_{x\to x_0} (f(x)\cdot g(x)) = y_0 \cdot z_0.$$

Proof. We apply Lemma 11.30. Let $x_k \to x_0,\ x_k \in E,\ x_k \neq x_0$. For $k \in \mathbb{N}$, let $u_k = f(x_k)$, $v_k = g(x_k)$. Then $u_k \to y_0$ and $v_k \to z_0$, so $u_k \cdot v_k \to y_0 \cdot z_0$ by Theorem 11.16(iii). ■

Exercises

11.6.6 Prove Theorem 11.37.

11.6.7 For those of you familiar with the vector cross product in $\mathbb{R}^3$ (see Section 11.10 for an introduction), does the analogue of Theorem 11.38 hold if $m = 3$ and the dot product is replaced by the cross product?

11.7 Continuity of Functions from $\mathbb{R}^n$ to $\mathbb{R}^m$

Now that we have the concept of limit for vector-valued functions, we can introduce the notion of continuity in just the same way we did for real functions in Section 5.4.2. The proofs in this section are virtually identical to the corresponding proofs in the one variable case and so we omit them.

Definition 11.39: Let $E \subset \mathbb{R}^n$, $f : E \to \mathbb{R}^m$, $x_0 \in E$. We say f is *continuous* at x_0 provided that for every $\varepsilon > 0$ there exists $\delta > 0$ such that $\|f(x) - f(x_0)\| < \varepsilon$ for all $x \in E$ for which $\|x - x_0\| < \delta$.

Observe that with this definition every function f is continuous at each isolated point of E, that is, any point of E that is not a limit point of E. For a limit point x_0 of E, it is easy to see that f is continuous at x_0 if and only if

$$\lim_{x \to x_0} f(x) = f(x_0).$$

Because of Lemma 11.30, a definition of continuity in terms of sequences is also immediately available: f is continuous at x_0 if and only if for every sequence $\{x_k\} \to x_0$ $(x_k \in E)$,

$$\lim_{k \to \infty} f(x_k) = f(x_0).$$

In terms of neighborhoods, we find that f is continuous at x_0 provided that for every neighborhood V of $f(x_0)$ there exists a neighborhood U of x_0 such that $f(U \cap E) \subset V$.

In short, all these definitions for continuity are equivalent and state in precise language that all points in E that are near x_0 map into points in $\mathbb{R}^m$ that are near $f(x_0)$.

Definition 11.40: Let $E \subset \mathbb{R}^n$ and let $f : E \to \mathbb{R}^m$. If f is continuous at all points of E, we say that f is *continuous on* E. When E is understood from the context we say simply that f is *continuous*.

A global characterization of continuity analogous to Theorem 5.37 for real functions on subsets of $\mathbb{R}$ is also available.

Theorem 11.41: *Let $E \subset \mathbb{R}^n$ and let $f : E \to \mathbb{R}^m$. Then f is continuous (on E) if and only if for every open set $V \subset \mathbb{R}^m$, the set*

$$f^{-1}(V) = \{x \in E : f(x) \in V\}$$

is open (relative to E). [This means that $f^{-1}(V)$ is the intersection of E with an open subset of $\mathbb{R}^n$.] In particular, if E is open, f is continuous if and only if for every open set $V \subset \mathbb{R}^m$, the set $f^{-1}(V)$ is open in $\mathbb{R}^n$.

The expected rules of continuity for sums, multiplication by a scalar, and dot products of functions are valid for vector-valued functions; their proofs involve no more than invoking the corresponding limit laws from Theorems 11.37 and 11.38.

From Theorem 11.36 we infer that a vector-valued function f is continuous at a point x_0 if and only if all of its coordinate functions f^i are continuous real-valued functions at x_0. This statement should not be confused with the *incorrect* statement that a real-valued function of several variables that is "continuous in each variable separately" is continuous.

Example 11.42: Consider once again Example 11.32,

$$f(x,y) = \frac{2xy}{x^2+y^2}\ ,\ f(0,0) = 0.$$

For each fixed $x = x_0$, $f(x_0, \cdot)$ is continuous on $\mathbb{R}$. The same is true of the functions $f(\cdot, y_0) : \mathbb{R} \to \mathbb{R}$. But f is not continuous at $(0,0)$ since

$$\lim_{(x,y) \to (0,0)} f(x,y)$$

does not exist. See also Exercises 11.7.12 and 11.7.13. ◀

Exercises

11.7.1 Prove that all the definitions for continuity of a function at a point are equivalent.

11.7.2 Show that continuity of f at x_0 does not depend on which of the norms $\|\cdot\|$, $\|\cdot\|_1$, $\|\cdot\|_\infty$ is in use.

11.7.3 Prove Theorem 11.41.

11.7.4 Let $E \subset \mathbb{R}^n$, $f, g : E \to \mathbb{R}^m$ and $\alpha \in \mathbb{R}$. Prove that if f and g are continuous at x_0, then so are $f+g$ and αf.

11.7.5 State carefully a theorem whose conclusion is that the composition of two continuous functions is continuous.

11.7.6 Let

$$f(x,y) = xy\left(\frac{x^2-y^2}{x^2+y^2}\right),\ f(0,0) = 7.$$

(This function was discussed in Example 11.31 of Section 11.6.) Is it possible to change the value of f at $(0,0)$ in such a way that the resulting function is continuous at $(0,0)$?

11.7.7 Let $f(r,\theta) = (r\cos\theta, r\sin\theta)$. (This function was discussed in Example 11.33 of Section 11.6.) Is f continuous at $(0,0)$? Is f continuous on all of $\mathbb{R}^2$?

11.7.8 Use Exercise 11.7.4 to prove that any polynomial of n-variables is continuous on $\mathbb{R}^n$. (See Exercise 11.5.1.)

11.7.9 Let $g(x_1,x_2,x_3) = \sqrt{1-x_1^2-x_2^2-x_3^2}$, where g is defined on the closed unit ball in $\mathbb{R}^3$. (This function was discussed in Example 11.21 of Section 11.5.) Is g continuous at $(1,0,0)$?

11.7.10 Prove that $\|\cdot\|$ is a continuous function on $\mathbb{R}^m$.

11.7.11 Prove that if $f : \mathbb{R}^n \to \mathbb{R}^m$ is continuous, then $\|f(x)\|$ is continuous on $\mathbb{R}^n$.
SEE NOTE 269

11.7.12 Let

$$f(x,y) = \frac{2x^2y}{x^4+y^2}$$

with $f(0,0) = 0$. Show

(a) The limit of $f(x,y)$ as $(x,y) \to (0,0)$ along any straight line is $f(0,0)$.

(b) f is discontinuous at $(0,0)$.

SEE NOTE 270

11.7.13 A careless student claims to have a proof of the *incorrect* statement,

> *If $f : \mathbb{R}^2 \to \mathbb{R}$ is continuous in each variable separately, then f is continuous.*

Proof: Let $(x_0,y_0) \in \mathbb{R}^2$. For $(x,y) \in \mathbb{R}^2$,

$$|f(x,y) - f(x_0,y_0)| \le |f(x,y) - f(x,y_0)| + |f(x,y_0) - f(x_0,y_0)|.$$

There exists δ_1 such that if $|y-y_0| < \delta_1$, then

$$|f(x,y) - f(x,y_0)| < \varepsilon/2$$

and δ_2 such that if $|x-x_0| < \delta_2$, then

$$|f(x,y_0) - f(x_0,y_0)| < \varepsilon/2.$$

Thus if $\delta = \min(\delta_1, \delta_2)$ and $|x-x_0| < \delta$, $|y-y_0| < \delta$, then

$$|f(x,y) - f(x_0,y_0)| < \varepsilon.$$

(a) What is the flaw in the "proof?"

(b) Find an added hypothesis that would make the quoted statement correct and the preceding outline of a proof valid.

(c) Show that if f is continuous in each variable separately on $\mathbb{R}^2$, then for every $(x_0,y_0) \in \mathbb{R}^2$,

$$\lim_{x\to x_0}(\lim_{y\to y_0} f(x,y)) \quad \text{and} \quad \lim_{y\to y_0}(\lim_{x\to x_0} f(x,y))$$

exist. Does this imply the existence of

$$\lim_{(x,y)\to(x_0,y_0)} f(x,y)?$$

SEE NOTE 271

11.8 Compact Sets in $\mathbb{R}^n$

In Section 4.5 we introduced the important notion of compactness for subsets of $\mathbb{R}$. In this section we extend this notion to subsets of $\mathbb{R}^n$. We shall see in the next section that properties of continuous functions defined on compact subsets of $\mathbb{R}$ extend to continuous functions defined on compact subsets of $\mathbb{R}^n$.

There are many equivalent definitions we can give for "compact set" in the setting of $\mathbb{R}^n$. (In Section 4.5 we saw that a set in $\mathbb{R}$ was compact if it was closed and bounded, or if it had the Bolzano-Weierstrass property, or if it had the Heine-Borel property.) Since we have already obtained the Bolzano-Weierstrass theorem (Theorem 11.19), it is natural to base our definition on the notion of sequences.

Definition 11.43: A set $E \subset \mathbb{R}^n$ is *compact* if every sequence in E has a convergent subsequence whose limit is in E.

Theorem 11.44: *A set $E \subset \mathbb{R}^n$ is compact if and only if E is closed and bounded.*

Proof. Suppose E is closed and bounded, and $\{x_k\}$ is a sequence of points in E. Since E is bounded, $\{x_k\}$ has a convergent subsequence $\{x_{k_j}\}$ by Theorem 11.19, Section 11.4. Let $x_0 = \lim_{j\to\infty} x_{k_j}$. Then x_0 is a limit point of E. Since E is closed, $x_0 \in E$ (by Exercise 11.3.1).

Conversely, suppose E is compact. We show that E is closed and bounded by contradiction. If E is not closed, then E has a limit point $x_0 \notin E$. This means there exists a sequence $\{x_k\}$ in E converging to x_0. Every subsequence of this sequence also converges to x_0, so there is no subsequence of $\{x_k\}$ converging *to a point of* E. Thus E must be closed.

If E is not bounded, there exists a sequence $\{x_k\}$ in E such that $\|x_k\| \geq k$ for all $k \in \mathbb{N}$. This sequence has no convergent subsequence. Thus E must be bounded. ■

Corollary 11.45: *A set $E \subset \mathbb{R}^n$ is closed and bounded if and only if every infinite subset of E has a point of accumulation that belongs to E.*

The proof follows immediately from Theorem 11.44. The Heine-Borel Theorem also carries over to $\mathbb{R}^n$. We leave this as Exercise 11.8.2.

Exercises

11.8.1 Prove Corollary 11.45.

11.8.2 ✂ Provide definitions for "open cover" and "Heine-Borel property" for subsets of $\mathbb{R}^n$. Your definitions should be consistent with the one-dimensional version in Section 4.5.4. State and prove the analogue of Theorem 4.33 for subsets of $\mathbb{R}^n$.

SEE NOTE 272

11.9 Continuous Functions on Compact Sets

We turn now to the behavior of continuous functions defined on compact sets of $\mathbb{R}^n$. There were three fundamental properties possessed by every continuous real-valued function defined on a compact subset of $\mathbb{R}$: The range of the function was bounded, the function attained its maximum and minimum values, and the continuity was uniform. Each of these extends to vector-valued functions with few complications.

Theorem 11.46: *If $E \subset \mathbb{R}^n$ is compact and $f : E \to \mathbb{R}^m$ is continuous on E, then the set $f(E)$ is compact.*

Proof. Let $\{y_k\}$ be a sequence in the set $f(E)$. We show that $\{y_k\}$ has a convergent subsequence with limit in $f(E)$. For each $k \in \mathbb{N}$ there exists $x_k \in E$ such that $f(x_k) = y_k$. Since E is compact, the sequence $\{x_k\}$ has a convergent subsequence $\{x_{k_j}\}$ converging to a point $x_0 \in E$. Since f is continuous at x_0, $f(x_0) = \lim_{j\to\infty} f(x_{k_j})$, and since $x_0 \in E$, $f(x_0) \in f(E)$.

The sequence $\{y_{k_j}\} = \{f(x_{k_j})\}$ is thus a convergent subsequence of $\{y_k\}$ that converges to the point $f(x_0) \in f(E)$. ■

Corollary 11.47: *Let $E \subset \mathbb{R}^n$ be compact and let $f : E \to \mathbb{R}$ be continuous. Then f achieves an absolute maximum and absolute minimum on E.*

Proof. The set $f(E)$ is a compact subset of $\mathbb{R}$ by Theorem 11.46. Thus $f(E)$ is closed and bounded by Theorem 11.44. Let $[c,d]$ be the smallest closed interval containing $f(E)$. Such an interval exists since $f(E)$ is bounded. Since $f(E)$ is closed, $c \in f(E)$ and $d \in f(E)$. The number c is the absolute minimum of f on E, and d is the absolute maximum of f on E. ■

Uniform Continuity Our definition of uniform continuity extends the concept from Section 5.6 to vector-valued functions.

Definition 11.48: Let $E \subset \mathbb{R}^n$ and let $f : E \to \mathbb{R}^m$. We say f is *uniformly continuous* (on E) if for every $\varepsilon > 0$ there exists $\delta > 0$ such that if $x, y \in E$ and $\|x - y\| < \delta$, then $\|f(x) - f(y)\| < \varepsilon$.

Theorem 11.49: *If $E \subset \mathbb{R}^n$ is compact and $f : E \to \mathbb{R}^m$ is continuous on E, then f is uniformly continuous on E.*

The proof is identical to the proof of Theorem 5.48 when norms replace absolute value signs.

Exercises

11.9.1 Prove that if $f : E \to \mathbb{R}^m$ is uniformly continuous on E, a subset of $\mathbb{R}^n$, and E is bounded, then f is bounded on E.

11.9.2 Prove Theorem 11.49 using the result of Exercise 11.8.2.

11.9.3 Show that the function $f(x) = \|x\|$ is uniformly continuous on $\mathbb{R}^n$.

11.9.4 Let $E \subset \mathbb{R}^n$ be compact and let $f : E \to \mathbb{R}^m$. Prove that if f is continuous on E and one-to-one, then f^{-1} is continuous on the set $f(E)$. Show that this conclusion might fail if E is not compact.

SEE NOTE 273

11.9.5 Let $S = [0,1] \times [0,1]$, $f : S \to \mathbb{R}$. Prove that if f is continuous on S and $g : [0,1] \to \mathbb{R}$ is defined by

$$g(x) = \max_{0 \le y \le 1} f(x,y),$$

then g is continuous on $[0,1]$.

11.9.6 Let E be a compact subset of $\mathbb{R}^n$ and let $x \in \mathbb{R}^n$. Let

$$\operatorname{dist} x = \inf\{\|x - y\| : y \in E\}.$$

(a) Prove that "inf" can be replaced by "min" in the definition of "dist."

(b) Prove that "dist" is a continuous function on $\mathbb{R}^n$.

(c) Are parts (a) and (b) valid if E is assumed closed but not bounded? What if E is assumed bounded but not closed?

11.10 Additional Remarks

✂ *Enrichment section. May be omitted.*

We mention in this section some items that we won't need in the next chapter but that may be of interest.

On Open and Closed Sets in $\mathbb{R}^n$ We saw in Chapter 4 that open sets in $\mathbb{R}$ have a particularly simple structure. Every nonempty open subset of $\mathbb{R}$ can be expressed as a finite or countably infinite union of pairwise disjoint open intervals (Theorem 4.15). Closed sets in $\mathbb{R}$ are relatively easy to visualize. They are made up of intervals, isolated points, limits of isolated points, etc., and Cantor sets.

In contrast, open or closed sets in $\mathbb{R}^n$, $n \geq 2$, can be more complicated. It is easy, for example, to think of a closed set that is the common boundary of two disjoint open sets—a circle will do the job. Think of the inside and the outside as the two disjoint open sets. Can you construct a closed set that is the common boundary of *three* pairwise disjoint open sets? Of five? Of infinitely many? Constructions of these types of sets are not easy, but they exist, and can arise in natural ways.[1]

The open sets must be very "wiggly" sets and their common boundary B is connected (in the sense introduced in Definition 11.50), yet B contains no arcs! Can you visualize such a set B?

The Four-Color Problem This famous problem[2] was finally settled affirmatively in 1976, after many mathematicians had tried in vain to solve it over a period of more than 100 years.

In loose language, the four-color problem, originally posed in 1852, asks whether "the regions of any map in the plane can be colored using no more than four colors, in such a way that those regions that have common boundaries consisting of more than one point have different colors." The solution that, in 1976, was finally announced for this problem remains controversial, however, since the proof required hundreds of pages and thousands of hours of computer verification.

Our preceding remarks about the boundaries of sets in higher-dimensional spaces shows the importance of being precise in stating problems. Suppose that $E_1, E_2, \ldots, E_5$ are bounded open sets with a common boundary B. Then five colors would be needed to color the resulting map. This would violate the four-color theorem as we loosely stated it. The actual four-color theorem is carefully stated, of course, so that the types of regions allowed and their common boundaries are limited appropriately.

Cross Products We have made considerable use of the dot product $x \cdot y$ of two vectors in $\mathbb{R}^n$. For the special case of $n = 3$, there is another important product, the *cross product* or *vector product*, denoted by $x \times y$. For any two vectors $x = (x_1, x_2, x_3)$ and $y = (y_1, y_2, y_3)$ we define

$$x \times y = (x_2y_3 - x_3y_2, x_3y_1 - x_1y_3, x_1y_2 - x_2y_1).$$

The cross product is important in various parts of mathematics and physics. We shall have no need of it in the chapters that follow, so we shall not provide a development here. We mention only that $x \times y$ is a vector that is perpendicular to the plane determined

[1] An interesting short discussion and original references can be found in J. H. Hubbard, The Forced Damped, Pendulum: Chaos, Complication and Control, *Amer. Math. Monthly*, **106**, No. 8 (Oct. 1999), pp. 745–747.

[2] An interesting book on the subject, written for a lay readership, is *The Four Color Theorem,* R. and G. Fritsch (Springer-Verlag, 1998).

by the vectors x and y, and

$$\|x \times y\| = \|x\| \, \|y\| \sin\theta,$$

where $0 \leq \theta \leq \pi$ is the angle determined by x and y. Geometrically, $\|x \times y\|$ is the area of the parallelogram of which x and y are adjacent sides.

Connectedness in $\mathbb{R}^n$ In Exercise 11.3.7 we defined connectedness for open sets in $\mathbb{R}^n$. An open set D in $\mathbb{R}^n$ is connected provided that each pair of points in D can be joined by a polygonal path lying in D. The notion of connectedness captures the idea of a set being in "one piece." What about sets that are not open? A closed disk, the graph of a continuous function $f : \mathbb{R} \to \mathbb{R}$, and the set

$$A = \{(x,y) : x^2 \leq y \leq 2x^2\}$$

are subsets of $\mathbb{R}^2$ that are not open but do seem to be in one piece. The definition involving polygonal paths applies to a closed disk and to the graphs of some, but not all, continuous functions. It does not apply to the set A since no polygonal path can join the origin to some other point of A. Here is a more general definition of connectedness in $\mathbb{R}^n$.

Definition 11.50: Let $S \subset \mathbb{R}^n$. If there exist disjoint open sets U and V such that $S \subset U \cup V$, $S \cap U \neq \emptyset$ and $S \cap V \neq \emptyset$, then U and V are said to *separate* S. The set S is *connected* if there are no open sets U, V that separate S.

With this definition, the previously mentioned sets are all connected. See the exercises for more examples and for consistency of this definition with the definition given earlier for open sets.

Exercises

11.10.1 Let D be an open set in $\mathbb{R}^n$. Prove that D is connected according to Definition 11.50 if and only if D is connected according to the definition in Exercise 11.3.7.

11.10.2 Verify that the sets (i) any closed disk, (ii) the graph of a continuous function $f : \mathbb{R} \to \mathbb{R}$ and (iii) the set $A = \{(x,y) : x^2 \leq y \leq 2x^2\}$ mentioned in our discussion of connectedness are all connected according to Definition 11.50.

11.10.3 Let $E \subset \mathbb{R}^n$, $f : E \to \mathbb{R}^m$, f continuous on E. Show that if E is connected, then $f(E)$ is connected.

11.10.4 State and prove an intermediate value property for continuous real-valued functions defined on connected sets.

11.10.5 Let $f : [a,b] \to \mathbb{R}$. Prove that f is continuous if and only if the graph of f is closed and connected (in $\mathbb{R}^2$).

11.10.6 Prove that the intersection of a descending sequence of compact connected sets in $\mathbb{R}^n$ is connected. Give an example to show that the statement is false if we drop the hypothesis of compactness.
SEE NOTE 274

11.10.7 Look at Hubbard's article cited in footnote 1. Observe that the common boundary of the open sets can be expressed as the intersection of a descending sequence of compact connected sets and is therefore connected by Exercise 11.10.6.

11.10.8 Verify the properties of the cross product $x \times y$ as we have defined it in this section. Show that $x \times y$ is orthogonal to x and to y.

Notes

[262]Exercise 11.1.5. See Definition 5.27 for the definition of the intermediate value property. The exercises in that section also provide a clue to the answer of this question.

[263]Exercise 11.1.6. For part (b) see Exercise 10.8.3.

[264]Exercise 11.2.7. Your answers to (b) and (c) illustrate a special feature of the euclidean norm. This norm relates to the dot product in certain useful ways. Our use of norms in this and the next chapter does not depend on this relationship, so we shall find it convenient to use $\|\cdot\|_1$ and $\|\cdot\|_\infty$ at certain times. See Chapter 13, Examples 13.7, 13.8, and 13.9 for more on this subject in a setting that involves infinite sequences in place of n-tuples. (A normed linear space is any vector space with a norm satisfying the conditions of Theorem 11.6. When that norm comes from a dot product via $\|x\|^2 = x \cdot x$, the space is called a *Hilbert space* and enjoys many special properties.)

[265]Exercise 11.2.8. In fact, the only norm on $\mathbb{R}^n$ for which this identity is valid is the euclidean norm.

[266]Exercise 11.3.7. For a notion of connectedness that is different from this polygonal arc definition and that applies to sets that need not be open, see the discussion in Section 11.10.

[267]Exercise 11.3.8. The point of this problem is that since the open sets are exactly the same, so too will be all the other concepts whose definitions can be given entirely in terms of open sets. The same will be true when in later sections we consider convergence of sequences or limits, continuity, and differentiability of functions.

[268]Exercise 11.6.5. Use the ideas in the proof of Lemma 11.34.

[269]Exercise 11.7.11. Use Exercise 11.7.10.

[270]Exercise 11.7.12. Show that $\lim_{t \to 0} f(t, t^2) = 1$.

[271]Exercise 11.7.13. For (a), look at the outline in conjunction with the example of Exercise 11.7.12. The hypothesis in (b) should involve an appropriate form of "uniform continuity in one of the variables with respect to the other variable." For (c) Example 11.32 of Section 11.6 and Exercise 11.7.12, illustrate that this does not imply that the double limit exists.

[272]Exercise 11.8.2. Section 13.12.3 establishes the equivalence of the Bolzano-Weierstrass and Heine-Borel properties in the more general setting of metric spaces.

[273]Exercise 11.9.4. For the counterexample you might use

$$f(x) = (\cos x, \sin x)$$

for $0 \leq x < 2\pi$.

[274]Exercise 11.10.6. Let

$$A_k = \{(0,0)\} \cup \{(1,0)\} \cup \{(x,y) : 0 < y < \frac{1}{k}\}.$$

Chapter 12

DIFFERENTIATION ON $\mathbb{R}^n$

12.1 Introduction

We shall assume that you are familiar with notions related to differentiation of functions of several variables. This familiarity should include some understanding of partial and directional derivatives, their roles in obtaining tangent lines and tangent planes when dealing with functions of two variables, and some comfort in performing calculations with partials.

Students in elementary calculus can usually master some of the simpler concepts related to differentiation of functions of several variables, but may have more difficulty understanding the meaning of differentiability (beyond that it implies existence of a tangent plane for $n = 2$). Concepts related to the differential or to various chain rules are often difficult to grasp.

In this chapter we study concepts related to differentiation of functions of several variables. We begin with fixing notation for partial and directional derivatives, and then proceed in a leisurely fashion to discuss a number of topics. When our experiences tell us that students find a topic difficult, we devote a section to "setting up" the material, leaving the proofs to the following section, or we begin with special cases, deferring the general situation until we believe you feel comfortable with the topic.

A case in point is the differential. If you are well-prepared, you can proceed rapidly to Section 12.8, browsing quickly through much of the earlier material. A much gentler approach is to use the earlier sections as a means for familiarizing yourself with the concept of differentiability, differential, and chain rule before attacking the more abstract setting of mappings from $\mathbb{R}^n$ to $\mathbb{R}^m$. We have labeled that section with ✂, indicating that it can be omitted, because it involves a bit of linear algebra—previous sections involve only a minimal use of linear algebra.

12.2 Partial and Directional Derivatives

For a function of two variables $f(x,y)$ there is an obvious first attempt at the study of derivatives. The founders of our subject (Newton, Leibniz, and others) carried calculus techniques over to such functions merely by differentiating according to the usual rules of calculus, treating each variable separately. The example and its notation should be familiar to the student from earlier courses.

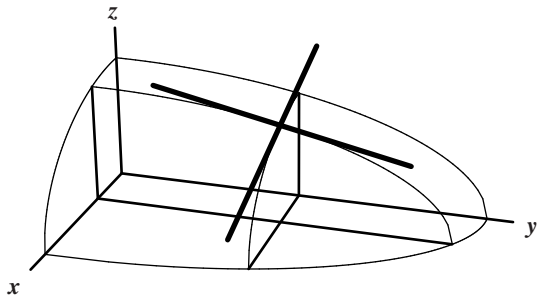


Figure 12.1. Tangent lines for the function $f(x,y) = \frac{1}{3}\sqrt{36-4x^2-y^2}$ at the point $(1,3,\sqrt{23}/3)$.

Example 12.1: Let $f(x,y) = (\sqrt{36-4x^2-y^2})/3$. You may recognize the surface corresponding to this function as half of an ellipsoid. The partial derivatives of f are

$$\frac{\partial}{\partial x}f(x,y) = \frac{-4x}{3\sqrt{36-4x^2-y^2}} \quad \text{and} \quad \frac{\partial}{\partial y}f(x,y) = \frac{-y}{3\sqrt{36-4x^2-y^2}}.$$

◀

A more formal way of saying what is happening here is to compare with the usual definition of the derivative of a function $f(x)$,

$$\frac{d}{dx}f(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}.$$

The process here for $f(x,y)$ is to perform the same computation but holding y as a fixed constant throughout:

$$\frac{\partial}{\partial x}f(x,y) = \lim_{h\to 0}\frac{f(x+h,y)-f(x,y)}{h}$$

and then holding x as a fixed constant,

$$\frac{\partial}{\partial y}f(x,y) = \lim_{h\to 0}\frac{f(x,y+h)-f(x,y)}{h}.$$

It is this that we will take as our definition for functions of two variables; for more variables the extension is similar. We shall prefer a subscript notation in place of the $\frac{\partial}{\partial x}$ notation when there are more than two or three variables.

The partials $f_1 = \frac{\partial f}{\partial x}$ and $f_2 = \frac{\partial f}{\partial y}$ at a point (x_0,y_0) provide slopes of two specific tangent lines to the surface $z = f(x,y)$. These are the lines in the planes $x = x_0$ and $y = y_0$ that go through the point (x_0,y_0,z_0) and are tangent to the surface. Figure 12.1 shows these tangent lines for the function of Example 12.1 at the point $(x_0,y_0,z_0) = (1,3,\sqrt{23}/3)$.

These first simple steps at a theory of differentiation for functions of two or more variables served the pioneers of the calculus well. As we shall see, however, a deeper look at differentiation is needed to advance much further.

12.2.1 Partial Derivatives

Suppose $f : D \to \mathbb{R}$, where D is an open subset of $\mathbb{R}^n$. Let

$$x = (x_1, \ldots, x_n) \in D.$$

If we allow one of the coordinates of x, say x_i, to vary while the others are fixed, we obtain a function of the one variable x_i. We may denote this function as

$$f(x_1, x_2, \ldots, x_{i-1}, \cdot, x_{i+1}, \ldots, x_n). \tag{1}$$

The dot in the ith position indicates that x_i may vary subject to the constraint that the point $(x_1, x_2, \ldots, x_{i-1}, x_i, x_{i+1}, \ldots, x_n)$ is in D.

If the derivative of the function (1) exists at x we obtain the *partial derivative* of f with respect to x_i at x. We denote this partial derivative at x by $f_i(x)$ or $f_i(x_1, \ldots, x_n)$. Thus

$$f_i(x) = \lim_{h \to 0} \frac{f(x_1, \ldots, x_i + h, \ldots, x_n) - f(x_1, \ldots, x_i, \ldots, x_n)}{h}. \tag{2}$$

Note. As was true about ordinary derivatives, it is true here too that different notations are sometimes useful, and many notations are in use. For example, $f_i(x)$ is often denoted by $\frac{\partial f}{\partial x_i}(x)$ or $f_{x_i}(x)$. If we write $u = f(x_1, \ldots, x_n)$ we often use the notations

$$u_i, \ \frac{\partial u}{\partial x_i}, \quad \text{or} \quad u_{x_i}$$

in place of

$$f_i, \ \frac{\partial f}{\partial x_i}, \quad \text{or} \quad f_{x_i}.$$

Of course, it is often convenient when we deal with only a few variables to denote these variables by x, y, z, instead of x_1, x_2, x_3.

Second-Order Partial Derivatives The partial derivatives f_i of a function $f(x)$ are themselves functions of x. If they are defined at all points of a neighborhood of x, they might themselves have partial derivatives at x. These are called the *second partial derivatives* of f at x or the partial derivatives of the *second order*.

There can be as many as n^2 second partials. For example, for $n = 2$ four second partials can be defined:

$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial x^2}, \qquad \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial y^2},$$
$$\frac{\partial}{\partial x}\left(\frac{\partial f}{\partial y}\right) = \frac{\partial^2 f}{\partial x \partial y}, \qquad \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right) = \frac{\partial^2 f}{\partial y \partial x}.$$

The two partials that involve both x and y are often called *cross partials* or *mixed partials* . We shall see presently that it is often, but not always, true that the mixed partials

$$\frac{\partial^2 f}{\partial y \partial x} \ \text{ and } \ \frac{\partial^2 f}{\partial x \partial y}$$

are equal. In any case be sure to grasp that they mean two different things and to sort out from the notation which derivative is performed first.

A typical second partial for a function f of n variables can be denoted as

$$\frac{\partial}{\partial x_i}\left(\frac{\partial f}{\partial x_j}\right) = \frac{\partial^2 f}{\partial x_i \partial x_j}.$$

Once again, note which derivation is being performed first.

What notation should we use for second partials if we write f_1 for $\frac{\partial f}{\partial x}$? It seems natural to write

$$f_{12} \text{ for } \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$$

because we would like f_{12} to be short for $(f_1)_2$. If we do so, then we have preserved order in one sense. But we have reversed it in another sense, since for f_{12} we compose from left to right ($(f_1)_2$) whereas in $\frac{\partial^2 f}{\partial y \partial x}$ we compose from right to left. Nonetheless, we shall use f_{12} to mean $(f_1)_2$ because it's a bit simpler to manipulate. In any case, there should be no confusion since there are few places in which both the "f_{ij}" notation and the "$\frac{\partial^2 f}{\partial x_i \partial x_j}$" notation are used simultaneously.

Note. We mention that some authors reverse the notation so that, for example, $\frac{\partial^2 f}{\partial x \partial y}$ means $\frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right)$. We prefer our notation because it preserves the order of composition, but you should be aware when reading material involving mixed partials (in other books) that the other convention might be in force.

Higher-Order Partial Derivatives We can also consider partial derivatives of higher orders. Thus, for example

$$\frac{\partial^3 f}{\partial x \partial y \partial y} \text{ would mean } \frac{\partial}{\partial x}\left(\frac{\partial^2 f}{\partial y^2}\right),$$

which we could also write as f_{221} meaning $(f_{22})_1$. As before the order in which the derivatives are taken is given by the notation and the convention must be remembered.

Exercises

12.2.1 Calculate $f_1(1,2\pi)$, $f_2(1,2\pi)$, $f_{12}(1,2\pi)$, and $f_{21}(1,2\pi)$ for the function

$$f(x,y) = x \sin y.$$

12.2.2 Let $u(x,y) = y^3 - 3x^2y$ and $v(x,y) = x^3 - 3xy^2$.

(a) Show that $\dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} = 0$ on $\mathbb{R}^2$.

(b) Show that $\dfrac{\partial^2 v}{\partial x^2} + \dfrac{\partial^2 v}{\partial y^2} = 0$ on $\mathbb{R}^2$.

(c) Show that $\dfrac{\partial u}{\partial x} = \dfrac{\partial v}{\partial y}$ and $\dfrac{\partial u}{\partial y} = -\dfrac{\partial v}{\partial x}$ on $\mathbb{R}^2$.

12.2.3 **(Harmonic Functions)** A function h defined on a region D of $\mathbb{R}^2$ is called *harmonic* if h has continuous partials of the first and second order and $h_{11} + h_{22} = 0$ at all points of D. This equation is called *Laplace's equation*. Verify that each of the following is harmonic on all of $\mathbb{R}^2$ and, for each, verify that $f_{12} = f_{21}$ and $f_{112} = f_{121} = f_{211}$.

(a) $e^x \cos y$

(b) $y^2 - 3x^2y$

(c) $x^2 - y^2 + 2y$

12.2.4 ✂ **(Cauchy-Riemann Equations)** The two equations introduced in Exercise 12.2.2(c)

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \text{ and } \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x} \tag{3}$$

are called the Cauchy-Riemann equations and are fundamental in complex analysis. This is so because a necessary and sufficient condition that a continuous function of a complex variable defined in a neighborhood N of a point in the complex plane be analytic is that its real and imaginary parts satisfy the Cauchy-Riemann equations. This means that if

$$f(x,y) = (u(x,y), v(x,y)),$$

then both equations in (3) are valid for all $(x,y) \in N$. [In complex notation we write $z = x + iy$ and $f(z) = u + iv$.] Show that each of the following functions $f = u + iv$ is analytic in a neighborhood of the origin.

(a) $u = x^2 - y^2$ and $v = 2xy$

(b) $u = e^x \cos y$ and $v = e^x \sin y$

(c) $u = 3x + y$ and $v = 3y - x$

12.2.5 Suppose u and v have all second-order partials on an open set $D \subset \mathbb{R}^2$ and that $v_{xy} = v_{yx}$ on D. Prove that if u and v satisfy the Cauchy-Riemann equations [equation (3) of Exercise 12.2.4] on D, then u is a *harmonic* function on D (see Exercise 12.2.3).

12.2.2 Directional Derivatives

Consider a function f of two variables. The partials f_1 and f_2 at a point (x_0, y_0) provide slopes of the two tangent lines to the surface in two directions. (Again see Figure 12.1.) There may be tangent lines corresponding to other vertical planes through (x_0, y_0, z_0), planes that are not parallel to the coordinate planes $x = 0$ or $y = 0$. The slopes of these tangent lines at (x_0, y_0, z_0) can be obtained by calculating *directional derivatives*.

Definition 12.2: Let $f : \mathbb{R}^2 \to \mathbb{R}$ and let $u = (u_1, u_2)$ be a unit vector in $\mathbb{R}^2$, that is,

$$\|u\| = \sqrt{u_1^2 + u_2^2} = 1.$$

The *directional* derivative of f in the direction u at the point (x_0, y_0) is

$$D_u f(x_0, y_0) = \lim_{t \to 0} \frac{f(x_0 + tu_1, y_0 + tu_2) - f(x_0, y_0)}{t} \tag{4}$$

if the limit exists.

Just as the partial derivatives measure the rate of change of f in the x and y directions, $D_u f(x_0, y_0)$ measures the rate of change of f in the direction of the vector u. Indeed, we should be aware that the partials *are themselves* directional derivatives: f_1 and f_2 are directional derivatives in the directions $(1,0)$ and $(0,1)$ and

$$f_1 = D_{(1,0)} \text{ and } f_2 = D_{(0,1)}.$$

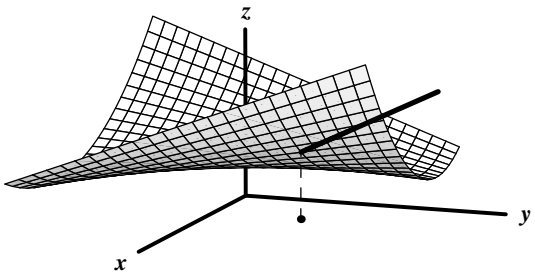


Figure 12.2. A directional derivative for $f(x,y) = x^2 + xy + 2$ at the point $(1,1)$ in the direction $(1/2, \sqrt{3}/2)$.

There is even a kind of converse: Later we will see that, under appropriate additional hypotheses, the derivatives in all directions can be written as soon as we know just the partials.

Observe that if $D_u f(x_0, y_0)$ exists, then so too does the derivative in the opposite direction $D_{-u} f(x_0, y_0)$, and

$$D_{-u} f(x_0, y_0) = -D_u f(x_0, y_0).$$

Example 12.3: Let $f(x,y) = x^2 + xy + 2$ and $u = (1/2, \sqrt{3}/2)$. To compute $D_u f(1,1)$ we calculate

$$\frac{f(1+\frac{1}{2}t, 1+\frac{\sqrt{3}}{2}t) - f(1,1)}{t} =$$
$$\frac{(\frac{3}{2}+\frac{\sqrt{3}}{2})t + (\frac{1}{4}+\frac{\sqrt{3}}{4})t^2}{t} \to \frac{3}{2} + \frac{\sqrt{3}}{2} \text{ as } t \to 0.$$

We shall see later that under suitable hypotheses, we can calculate

$$D_u f = (f_1, f_2) \cdot (u_1, u_2) = f_1 u_1 + f_2 u_2.$$

In this case $f_1(1,1) = 3$, $f_2(1,1) = 1$, so

$$(3,1) \cdot (\tfrac{1}{2}, \tfrac{\sqrt{3}}{2}) = \tfrac{3}{2} + \tfrac{\sqrt{3}}{2}$$

as we calculated before. Figure 12.2 illustrates the graph of f and shows the tangent line in the direction $(1/2, \sqrt{3}/2)$ at the point $(1,1)$. ◀

Exercises

12.2.6 Define directional derivatives for real valued functions of n variables.

SEE NOTE 275

12.2.7 Verify that f_1 and f_2 are directional derivatives in the directions $(1,0)$ and $(0,1)$ respectively (i.e., that $f_1 = D_{(1,0)}$ and $f_2 = D_{(0,1)}$). Check that $D_{-u} f(x_0, y_0) = -D_u f(x_0, y_0)$.

12.2.3 Cross Partials

Let f be defined on a region $D \subset \mathbb{R}^2$, and suppose both partial derivatives

$$\frac{\partial f}{\partial x}(x,y) \quad \text{and} \quad \frac{\partial f}{\partial y}(x,y)$$

exist at each point (x,y) in D. These partial derivatives are themselves functions in D and, as such, might in turn possess partial derivatives. In this case, there are four possible partial derivatives:

$$\frac{\partial^2 f}{\partial x^2} = \frac{\partial^2 f}{\partial x \partial x}\ ,\ \frac{\partial^2 f}{\partial y \partial x}\ ,\ \frac{\partial^2 f}{\partial x \partial y} \text{ and } \frac{\partial^2 f}{\partial y \partial y} = \frac{\partial^2 f}{\partial y^2}.$$

We recall, for example, that our notation requires

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial}{\partial y}\left(\frac{\partial f}{\partial x}\right).$$

The two partials $\frac{\partial^2 f}{\partial y \partial x}$ and $\frac{\partial^2 f}{\partial x \partial y}$ are often called *cross partials* or *mixed partials*.

Students in elementary calculus frequently believe there are only three second-order partials because it is usually true that

$$\frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}. \tag{5}$$

In this section we determine conditions under which (5) is valid. We begin with an example that illustrates that (5) is not always valid.

Example 12.4: Let

$$f(x,y) = \begin{cases} \frac{xy(x^2-y^2)}{x^2+y^2}, & \text{if } (x,y) \neq (0,0) \\ 0, & \text{if } (x,y) = (0,0) \end{cases}.$$

Using the subscript notation, we compute the partial derivatives

$$f_1 = \frac{\partial f}{\partial x}\ ,\ f_2 = \frac{\partial f}{\partial y}\ ,\ f_{12} = \frac{\partial^2 f}{\partial y \partial x}\ ,\ f_{21} = \frac{\partial^2 f}{\partial x \partial y}$$

at the point $(0,0)$. Now

$$f_1(0,y) = \lim_{h\to 0} \frac{f(h,y) - f(0,y)}{h} = \lim_{h\to 0} \frac{\left(\frac{hy(h^2-y^2)}{h^2+y^2}\right)}{h} = -y.$$

Similarly, $f_2(x,0) = x$. It follows that

$$\begin{aligned} f_{12}(0,0) &= \lim_{k\to 0} \frac{f_1(0,k) - f_1(0,0)}{k} = \frac{-k-0}{k} = -1 \text{ and} \\ f_{21}(0,0) &= \lim_{h\to 0} \frac{f_2(h,0) - f_2(0,0)}{h} = \lim_{h\to 0} \frac{h-0}{h} = 1. \end{aligned}$$

Thus the two cross partials are not equal. ◀

What is it that "went wrong" here? Consider the two identical expressions involving the function f of Example (12.4):

$$\frac{1}{k}\left[\frac{f(h,k)-f(0,k)}{h}-\frac{f(h,0)-f(0,0)}{h}\right], \tag{6}$$

$$\frac{1}{h}\left[\frac{f(h,k)-f(h,0)}{k}-\frac{f(0,k)-f(0,0)}{k}\right]. \tag{7}$$

If we first let $h \to 0$ in (6), and then let $k \to 0$, we obtain $f_{12}(0,0)$. On the other hand, letting $k \to 0$ in (7) and then letting $h \to 0$, we obtain $f_{21}(0,0)$. As we saw, these two iterated limits were not equal.

Note. You might recall that we have seen this phenomenon before, that two limits might be different if performed in different orders. In Example 9.7 we saw that the two iterated limits of a double sequence $\{s_{nm}\}$ might not be equal. See also Exercise 9.2.8 and its hint for a discussion of this phenomenon. Here we once again have a situation that involves iterated limits.

We can explain now what went wrong in our example: The failure of the two mixed partial derivatives f_{12} and f_{21} to be equal can be traced to the fact that the limit of these expressions (6) or (7) as $(h,k) \to 0$ (i.e., the *double limit*) fails to exist.

A simple sufficient condition for the existence of this double limit appears in the following theorem.

Theorem 12.5: *Let f be defined in a neighborhood of $(x_0,y_0) \in \mathbb{R}^2$. Suppose f has partial derivatives f_1, f_2, f_{12} and f_{21} in this neighborhood and that the mixed partials f_{12} and f_{21} are continuous at (x_0,y_0). Then $f_{12}(x_0,y_0) = f_{21}(x_0,y_0)$.*

Proof. Consider the two equal expressions

$$\frac{1}{k}\left[\frac{f(x_0+h,y_0+k)-f(x_0,y_0+k)}{h}-\frac{f(x_0+h,y_0)-f(x_0,y_0)}{h}\right] \tag{8}$$

$$\frac{1}{h}\left[\frac{f(x_0+h,y_0+k)-f(x_0+h,y_0)}{k}-\frac{f(x_0,y_0+k)-f(x_0,y_0)}{k}\right] \tag{9}$$

If we first let $h \to 0$ in (8) and then let $k \to 0$, we obtain $f_{12}(x_0,y_0)$. On the other hand, letting $k \to 0$ and then $h \to 0$ in (9), we obtain $f_{21}(x_0,y_0)$. We shall show that these two iterated limits are equal.

Observe first that the numerator D_{hk} in brackets in expression (8) can be written in the form

$$(f(x_0+h,y_0+k)-f(x_0+h,y_0))-(f(x_0,y_0+k)-f(x_0,y_0)). \tag{10}$$

Applying the mean value theorem (Theorem 7.20) to (10), we obtain

$$D_{hk} = (f_1(\xi,y_0+k)-f_1(\xi,y_0))h, \tag{11}$$

where ξ is between x_0 and x_0+h. Applying the mean value theorem again, we obtain

$$D_{hk} = f_{12}(\xi,\zeta)hk,$$

where ζ is between y_0 and y_0+k.

We can rewrite (10) in the form

$$(f(x_0+h,y_0+k)-f(x_0,y_0+k))-(f(x_0+h,y_0)-f(x_0,y_0)).$$

Applying the same argument as before, we obtain

$$D_{hk}=f_{21}(\sigma,\tau)kh,$$

where σ is between x_0 and x_0+h and τ is between y_0 and y_0+k.

Now let $(h,k)\to(0,0)$. Then $(\xi,\zeta)\to(x_0,y_0)$ and $(\sigma,\tau)\to(x_0,y_0)$. Since the functions f_{12} and f_{21} are continuous at (x_0,y_0) (by hypothesis), we obtain

$$\lim_{(h,k)\to(0,0)}\frac{D_{hk}}{hk}=f_{21}(x_0,y_0)=f_{12}(x_0,y_0)$$

as required. ■

Note. Observe that our proof actually shows that the *double limit*

$$\lim_{(h,k)\to(0,0)}\frac{D_{hk}}{hk}=f_{21}(x_0,y_0)=f_{12}(x_0,y_0).$$

This is a stronger statement than the conclusion of the theorem. We will return to this idea later when we discuss differentiability.

Theorem 12.5 shows that continuity of the partials is a sufficient condition for f_{12} to equal f_{21}. This condition is often met but is not a necessary condition.

Example 12.6: We can construct an example by using the function

$$g(x)=x^2\sin\frac{1}{x},\qquad g(0)=0,$$

which is differentiable on $\mathbb{R}$ but whose derivative g' is discontinuous at $x=0$. It follows that the function

$$f(x,y)=yx^2\sin\frac{1}{x},\qquad f(0,y)=0$$

has both partials f_{12} and f_{21} discontinuous at $(0,0)$. Thus Theorem 12.5 does not apply here, yet the partials

$$f_{21}(0,0)=f_{12}(0,0)=0$$

are equal. ◄

Other sufficient conditions for the equality of the mixed partials are known. For example, it suffices to assume existence of both mixed partials and continuity of only one of them at (x_0,y_0) or to assume that f_1 and f_2 are differentiable[1] at (x_0,y_0). We omit the proofs.

Theorem 12.5 extends readily to higher-order derivatives. For example, if f has partial derivatives of order one, two, and three, and all of these are continuous at (x_0,y_0), then

$$f_{221}=(f_2)_{21}=(f_2)_{12}=f_{212}=(f_{21})_2=(f_{12})_2=f_{122}.$$

[1] Differentiable in the sense defined later in Section 12.4.

Partial Derivatives of $f : \mathbb{R}^n \to \mathbb{R}$ Theorem 12.5 also extends to functions of more than two variables. The following theorem reduces to Theorem 12.5 by holding all of the variables except the ith and jth fixed in the proof. Only the notation becomes more complicated.

Theorem 12.7: *Let f be defined in some neighborhood of the point $\mathbf{x_0}$ in $\mathbb{R}^n$. Suppose f has partial derivatives f_i, f_j, f_{ij}, and f_{ji} in this neighborhood and that the mixed partials f_{ij} and f_{ji} are continuous at $\mathbf{x_0}$; then $f_{ij}(\mathbf{x_0}) = f_{ji}(\mathbf{x_0})$.*

Exercises

12.2.8 In general how many mixed partials of the third order does a function $f : \mathbb{R}^2 \to \mathbb{R}$ have? How many for a function $f : \mathbb{R}^n \to \mathbb{R}$?

12.2.9 For each of the harmonic functions of Exercise 12.2.3

(a) $f(x,y) = e^x \cos y$
(b) $f(x,y) = y^2 - 3x^2y$
(c) $f(x,y) = x^2 - y^2 + 2y$

verify that $f_{12} = f_{21}$ and $f_{112} = f_{121} = f_{211}$.

12.2.10 Let $f(x,y) = x^2 \tan^{-1}(y/x) - y^2 \tan^{-1}(x/y)$ when $x \neq 0$ and $y \neq 0$, and $f(x,y) = 0$ if either x or y is zero. Compute $f_{12}(0,0)$ and $f_{21}(0,0)$. Does your result contradict Theorem 12.5?

12.2.11 **(Double limits and iterated limits, revisited)** A bit of care is needed with the statement "If a double limit exists, so do the two iterated limits, and the two iterated limits equal the double limit." Let

$$f(x,y) = \begin{cases} y + x \sin \frac{1}{y} & \text{if } y \neq 0 \\ 0 & \text{if } y = 0 \end{cases}.$$

(a) Show that $\lim_{(x,y)\to(0,0)} f(x,y) = 0$ but $\lim_{x\to 0} \lim_{y\to 0} f(x,y)$ does not exist.

(b) Prove that if $\lim_{(x,y)\to(x_0,y_0)} g(x,y)$ and $\lim_{y\to y_o} g(x,y)$ exist for all x, then

$$\lim_{x\to x_0} \lim_{y\to y_o} g(x,y) = \lim_{(x,y)\to(x_0,y_0)} g(x,y).$$

12.3 Integrals Depending on a Parameter

Let g be continuous on $[a,b]$ and let

$$G(y) = \int_a^y g(x)\, dx. \tag{12}$$

We recall from the fundamental theorem of calculus that $G'(y) = g(y)$ for all $y \in [a,b]$. This represents half of the inverse nature of differentiation and integration.

Let us now complicate things a bit. Let f be continuous on a rectangle

$$R = \{(x,y) \in \mathbb{R}^2 : a \leq x \leq b, c \leq y \leq d\}.$$

Define a function $F : [c,d] \to \mathbb{R}$ by

$$F(y) = \int_a^b f(x,y)\, dx. \tag{13}$$

To see that F is well defined, we need only note that for every fixed $y \in [c,d]$, $f(\cdot,y)$ is simply a continuous function on $[a,b]$ and therefore the integral does exist.

We might consider (13) as a "partial integral," in the same way in which we consider our derivatives as partial derivatives: In both cases an operation is performed on one variable, while the other variable is held fixed.

What can we say about the function F? Is F continuous on $[c,d]$? Is F differentiable? If so, how can we compute F'? We address such questions in this section.

Observe two differences between (12) and (13). In (12) we are dealing with a function g of *one* variable; in (13) f is a function of two variables. Furthermore, in (12) the upper limit is the variable y, whereas in (13) the upper limit is the constant b.

Integrals of the type $\int_a^b f(x,y)\,dx$ appear frequently in practice. We think of y as a parameter and we are often concerned with the question "How does a small change in the parameter affect the resulting integral?" We address this question first with respect to continuity (of F) and then differentiability.

Theorem 12.8: *If f is continuous on $R = [a,b] \times [c,d]$ and*

$$F(y) = \int_a^b f(x,y)\,dx,$$

then F is continuous on $[c,d]$.

Proof. Let $y_0 \in [c,d]$. We prove F is continuous at y_0. Since R is closed and bounded, and therefore compact, f is uniformly continuous on R by Theorem 11.49. Thus, for $\varepsilon > 0$ there exists $\delta > 0$ such that for each $x \in [a,b]$

$$|f(x,y) - f(x,y_0)| < \frac{\varepsilon}{b-a}$$

whenever $|y - y_0| < \delta$. For $|y - y_0| < \delta$ we calculate

$$\begin{aligned} |F(y) - F(y_0)| &= \left| \int_a^b (f(x,y) - f(x,y_0))\,dx \right| \\ &\leq \int_a^b |f(x,y) - f(x,y_0)|\,dx \leq \int_a^b \frac{\varepsilon}{b-a}\,dx = \varepsilon. \end{aligned}$$

Thus $\lim_{y \to y_0}(F(y) - F(y_0)) = 0$ and F is continuous at y_0. ■

We turn now to the question of differentiability of F. We would like to find a formula that obtains F' via a "differentiation under the integral sign." But here f is a function of two variables, and a partial derivative is called for. Theorem 12.9 is often called Leibniz's rule.

Theorem 12.9 (Leibniz's Rule) *Let f be continuous on the rectangle*

$$R = [a,b] \times [c,d]$$

and let

$$F(y) = \int_a^b f(x,y)\,dx$$

for each $y \in [c,d]$. If the partial derivative $\frac{\partial f}{\partial y}$ exists and is continuous on R, then F is differentiable on $[c,d]$, and

$$F'(y) = \int_a^b \frac{\partial}{\partial y} f(x,y)\,dx.$$

Proof. It suffices to show that for each $y \in [c,d]$

$$\lim_{h\to 0}\left(\frac{F(y+h)-F(y)}{h} - \int_a^b \frac{\partial}{\partial y} f(x,y)\,dx\right) = 0. \tag{14}$$

Now for $y+h \in [c,d]$,

$$F(y+h)-F(y) = \int_a^b [f(x,y+h)-f(x,y)]\,dx. \tag{15}$$

It follows from the mean value theorem (Theorem 7.20) that for fixed x, y, and h, there exists a number $\xi \in (0,1)$ such that

$$f(x,y+h)-f(x,y) = h\frac{\partial}{\partial y} f(x,y+\xi h). \tag{16}$$

Substituting (16) in (15), we obtain

$$\frac{F(y+h)-F(y)}{h} - \int_a^b \frac{\partial}{\partial y} f(x,y)\,dx = \tag{17}$$
$$\int_a^b \left(\frac{\partial}{\partial y} f(x,y+\xi h) - \frac{\partial}{\partial y} f(x,y)\right) dx.$$

Since $\frac{\partial f}{\partial y}$ is continuous (by assumption), it is uniformly continuous on the compact set R. Thus, for $\varepsilon > 0$ there exists $\delta > 0$ such that if $(x,y) \in R$ and $(x',y') \in R$ with $|x-x'| < \delta$ and $|y-y'| < \delta$, then

$$\left|\frac{\partial}{\partial y} f(x,y) - \frac{\partial}{\partial y} f(x',y')\right| < \frac{\varepsilon}{b-a}.$$

In particular, if $|h| < \delta$, then

$$\left|\frac{\partial}{\partial y} f(x,y+\xi h) - \frac{\partial}{\partial y} f(x,y)\right| < \frac{\varepsilon}{b-a}. \tag{18}$$

From (17) and (18) we infer that for $|h| < \delta$,

$$\left|\frac{F(y+h)-F(y)}{h} - \int_a^b \frac{\partial}{\partial y} f(x,y)\,dx\right| \le \int_a^b \frac{\varepsilon}{b-a}\,dx = \varepsilon$$

and (14) follows. ■

Example 12.10: If

$$F(y) = \int_a^b \sin(xy)\,dx,$$

we can conclude from Theorem 12.9 that

$$F'(y) = \int_a^b x\cos(xy)\,dx.$$

We simply differentiate "through" the integral sign. ◄

In this case, both integrals appearing can easily be integrated by elementary methods (and Leibniz's rule thus verified), but this need not be true in general. It is often not possible to evaluate one or both of the integrals

$$\int_a^b \frac{\partial}{\partial y} f(x,y)\,dx \quad \text{or} \quad \int_a^b f(x,y)\,dx$$

in terms of elementary functions. We continue with some examples of Leibniz's rule used in conjunction with other techniques.

Example 12.11: To compute the derivative of the function

$$F(y) = \int_0^{\pi/2} \frac{dx}{\sqrt{1-y^2\sin^2 x}}$$

for $0 < y < 1$ we simply apply Leibniz's rule to obtain

$$F'(y) = \int_0^{\pi/2} \frac{y\sin^2 x}{\sqrt{1-y^2\sin^2 x}}\,dx.$$

This is an example of an elliptic integral with parameter y. ◀

Example 12.12: The integral

$$F(y) = \int_0^{y^2} yx^3\,dx \tag{19}$$

and its derivative can be calculated easily by elementary means. As an exercise, we calculate F' using Leibniz's rule. We observe first that there is now a variable upper limit of integration, so more than a direct application of Leibniz's rule is required.

In this case, the parameter occurs in a limit of integration as well as in the integrand, so we need to use the fundamental theorem of calculus and the chain rule as well as Leibniz's rule. (The chain rule is proved in Theorem 12.34 of Section 12.5.4.)

Let

$$u = y^2, \quad v = y, \text{ and } G(u,v) = \int_0^u vx^3\,dx.$$

Then the fundamental theorem of calculus gives us

$$\frac{\partial G}{\partial u} = vu^3$$

and Leibniz's rule provides

$$\frac{\partial G}{\partial v} = \int_0^u x^3\,dx = \frac{u^4}{4}$$

and the chain rule that we remember from the calculus of functions of two variables gives us

$$\frac{d}{dy}F(y) = \frac{\partial G}{\partial u}\frac{du}{dy} + \frac{\partial G}{\partial v}\frac{dv}{dy}$$

$$= vu^3 2y + \frac{u^4}{4}\cdot 1 = 2y^8 + \frac{y^8}{4} = \frac{9}{4}y^8,$$

which is the same result we would obtain by evaluating the integral (19) and differentiating. ◀

Some of the exercises involve the use of chain rules from calculus. These and other chain rules are discussed in greater detail and proved in Sections 12.5, 12.5.4, 12.5.6, and 12.8.

Exercises

12.3.1 Use methods of elementary calculus to verify directly the result in Example 12.12.

12.3.2 Calculate $F'(y)$ for each of the following functions F.

(a) $F(y) = \displaystyle\int_0^1 \mathrm{e}^{-x^2y^2}\,dx$

(b) $F(y) = \displaystyle\int_0^y \mathrm{e}^{-x^2y^2}\,dx$

(c) $F(y) = \displaystyle\int_0^{y^3} \mathrm{e}^{-x^2y^2}\,dx$

12.3.3 Let u and v have continuous derivatives on $\mathbb{R}$ and let f be continuous with continuous partial derivatives on $\mathbb{R}^2$. Obtain a formula for $F'(y)$, where

$$F(y) = \int_{u(y)}^{v(y)} f(x,y)\,dx.$$

SEE NOTE 276

12.3.4 The integral

$$F(y) = \int_0^y \sqrt{1-k^2\sin^2 t}\,dt$$

is called an *elliptic integral of the second kind*. (It arises in computing the length of an arc of an ellipse given parametrically by $x = a\cos t$, $y = b\sin t$, $0 < a < b$.) Calculate $F'(y)$.

12.3.5 Using elementary calculus techniques, show that the arc length of an ellipse given parametrically by

$$x = a\cos t,\ y = b\sin t, \quad (0 < a < b)$$

is given by the integral

$$b\int_0^{2\pi} \sqrt{1-k^2\sin^2 t}\,dt \ \text{ where } k^2 = \frac{b^2-a^2}{b^2}.$$

12.4 Differentiable Functions

When a function f of one real variable has a finite derivative at a point x_0, we say that f is differentiable at x_0. This is simply the definition of differentiability. Differentiable functions of one variable have many useful properties. The most fundamental of these is that the tangent line at x_0 is a close approximation to the function near x_0.

We might be tempted to carry this language over to higher dimensions and say that a function of several variables that has finite partial derivatives with respect to each variable is "differentiable." But such a definition would not be useful. It would not, moreover, generalize this notion that the tangent line is a close approximation to a differentiable function.

12.4.1 Approximation by Linear Functions

Let's see what is involved by looking at the one variable situation more carefully. Suppose $f:\mathbb{R}\to\mathbb{R}$, $x_0\in\mathbb{R}$, and

$$L(x)=a_0x+a_1$$

is *any* affine function[2] such that $L(x_0)=f(x_0)$. If f is continuous at x_0, then for x near x_0 we will have $f(x)$ near $L(x)$. More precisely,

$$\lim_{x\to x_0}|f(x)-L(x)|=0. \tag{20}$$

We could say L approximates f near x_0 in the sense of (20). Thus any affine function through the point $(x_0,f(x_0))$ approximates f near x_0 if f is continuous there.

If f is differentiable at x_0, we can obtain a better approximation using the tangent line T. This line has the equation

$$T(x)=f(x_0)+f'(x_0)(x-x_0).$$

Now, differentiability of f at x_0 means that

$$\left|\frac{f(x)-f(x_0)}{x-x_0}-f'(x_0)\right|\to 0 \text{ as } x\to x_0. \tag{21}$$

We can write (21) in the form

$$\left|\frac{f(x)-T(x)}{x-x_0}\right|\to 0 \text{ as } x\to x_0. \tag{22}$$

Comparing (22) with (20), we see the improvement in the approximation when we use the tangent line (when it exists): In (20) we divide $|f(x)-L(x)|$ by the constant 1 and obtain the limit 0; in (22) we divide $|f(x)-T(x)|$ by $|x-x_0|$, which approaches 0 as $x\to x_0$, and still obtain the limit 0. Thus not only is $T(x)$ near $f(x)$ when x is near x_0, but the distance between $T(x)$ and $f(x)$ is small in comparison with the distance between x and x_0.

Finally, we rewrite (22) in a form convenient for generalization to functions of several variables. If we write $h=x-x_0$ and manipulate (22) a bit, we arrive at the equivalent formulation

$$f(x_0+h)-f(x_0)=f'(x_0)h+\varepsilon h, \tag{23}$$

where $\varepsilon\to 0$ as $h\to 0$. Here ε depends on h [to satisfy the equality in (23)], and the requirement for differentiability (21) is that $\varepsilon\to 0$ as $h\to 0$.

12.4.2 Definition of Differentiability

We shall imitate the preceding for functions of several variables. To keep notation simple for the moment, let us begin by considering functions of two variables defined

[2] A real function L of a real variable is said to be *affine* if its graph is a straight line. Thus $L(x)=a_0x+a_1$. In calculus courses such functions are usually said to be "linear," but our language must be rather more precise since we are using some linear algebra in our presentation. Note that it is the function $f(x)-f(x_0)$ that can be approximated by a *linear* function a_0x.

in a neighborhood of a point $(x_0, y_0) \in \mathbb{R}^2$. We wish to express the condition that the surface corresponding to the equation

$$z = f(x,y)$$

has a tangent plane at (x_0, y_0). Thus the tangent plane T replaces the tangent line for functions of one variable.

Recall from elementary calculus that the tangent plane, if it exists, can be expressed by the linear equation

$$T(x,y) = f(x_0,y_0) + f_1(x_0,y_0)(x-x_0) + f_2(x_0,y_0)(y-y_0).$$

In order for this plane to be the tangent plane, we require that the analogue of (22) be valid, namely that

$$\frac{|f(x,y) - T(x,y)|}{\sqrt{(x-x_0)^2 + (y-y_0)^2}} \to 0 \text{ as } (x,y) \to (x_0,y_0). \tag{24}$$

Manipulating (24) as we did (22) and writing $h = x - x_0$, $k = y - y_0$, we arrive at the correct formulation for the two-variable version.

Definition 12.13: A function $f : \mathbb{R}^2 \to \mathbb{R}$ is *differentiable* at (x_0, y_0) if

1. The partial derivatives f_1 and f_2 exist at (x_0, y_0).
2. It is possible to write

$$\begin{aligned} f(x_0+h, y_0+k) - f(x_0,y_0) = \\ f_1(x_0,y_0)h + f_2(x_0,y_0)k + \varepsilon(h,k)\sqrt{h^2+k^2}, \end{aligned} \tag{25}$$

   where $\varepsilon(h,k) \to 0$ as $\sqrt{h^2+k^2} \to 0$ (with $\varepsilon(0,0) = 0$).

Note. By observing that for $(h,k) \in \mathbb{R}^2$,

$$\sqrt{h^2+k^2} \le |h| + |k| \le \sqrt{2}\sqrt{h^2+k^2},$$

we can rewrite statement (2) in the definition in the simpler form

2′. It is possible to write

$$\begin{aligned} f(x_0+h, y_0+k) - f(x_0,y_0) = \\ f_1(x_0,y_0)h + f_2(x_0,y_0)k + \varepsilon(h,k)(|h| + |k|), \end{aligned} \tag{26}$$

where $\varepsilon(h,k) \to 0$ as $|h| + |k| \to 0$.

The condition $|h| + |k| \to 0$ is equivalent to $h \to 0$ and $k \to 0$, which is, in turn, equivalent to $\sqrt{h^2+k^2} \to 0$.

Let us return for a moment to the elementary geometry underlying the role of the tangent plane. This plane should approximate f near (x_0, y_0) in the sense of (24). Roughly, this requires that when (x,y) is near (x_0,y_0), then $|f(x,y) - T(x,y)|$ is small in comparison with the distance between (x,y) and (x_0,y_0). Mere existence of the partials at (x_0,y_0) does *not* guarantee this; that would guarantee only that the desired comparison is small when (x,y) is near (x_0,y_0) and $x = x_0$ (or $y = y_0$). Appropriate tangent lines exist in the intersection of the surface $z = f(x,y)$ with the plane $x = x_0$ (or $y = y_0$), but (24) need not hold when $(x,y) \to (x_0,y_0)$ in other ways.

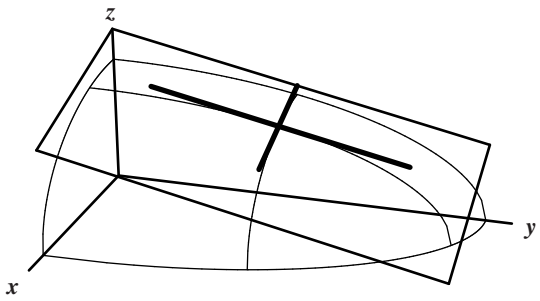


Figure 12.3. Tangent plane for the function $f(x,y) = \frac{1}{3}\sqrt{36-4x^2-y^2}$ at the point $(1,3,\sqrt{23}/3)$.

Example 12.14: Consider again the function

$$f(x,y) = (\sqrt{36-4x^2-y^2})/3$$

from Example 12.1. In Figure 12.1 we illustrated the tangent lines at the point $(x_0,y_0,z_0) = (1,3,\sqrt{23}/3)$. With some more effort we could now show that this function is differentiable at that point by going through the computations in the definition to show that the tangent plane approximates the function in the correct sense. Figure 12.3 shows this tangent plane and illustrates that this approximation is plausible. In practice we would normally apply some theorem (such as Theorem 12.21) that would allow us to conclude differentiability without resorting to such computations. ◀

Example 12.15: Let $f(x,y) = \sqrt{|xy|}$. (This function is reminiscent of the function $|x| = \sqrt{|x^2|}$, which is continuous but not differentiable at $x = 0$.) It is clear that f is continuous at $(0,0)$ (Exercise 12.4.1). Since $f = 0$ on the coordinate axes,

$$f_1(0,0) = f_2(0,0) = 0. \tag{27}$$

For f to be differentiable at $(0,0)$ it must be true that $\varepsilon(h,k) \to 0$ as $|h|+|k| \to 0$, where $\varepsilon(h,k)$ is defined by

$$\sqrt{|hk|} = f_1(0,0)h + f_2(0,0)k + \varepsilon(h,k)(|h|+|k|).$$

By (27), $\sqrt{|hk|} = \varepsilon(|h|+|k|)$.

Letting $h = -k$, for example, we obtain $|h| = 2\varepsilon|h|$, so $\varepsilon(h,-h) = \frac{1}{2}$. Thus it is not the case that $\varepsilon(h,k) \to 0$ as $|h|+|k| \to 0$. (Letting $h = -k$ is equivalent to having (x,y) approach (x_0,y_0) along the line $x+y=0$, $z=0$.) Figure 12.4 illustrates the graph of the function f, evidently shaped like a folded napkin with the folds along the two axes. Note especially that the information along the "folds" that $f_1(0,0) = f_2(0,0) = 0$ does not help to compute the directional derivative in the direction $(-1/\sqrt{2}, 1/\sqrt{2})$ (illustrated in the figure) and that corresponds to the case $h = -k$ discussed previously. ◀

Differentiability for $f : \mathbb{R}^n \to \mathbb{R}$. With these preliminaries we can now give a formal definition for *differentiability* of a function $f : \mathbb{R}^n \to \mathbb{R}$. This is the obvious generaliza-

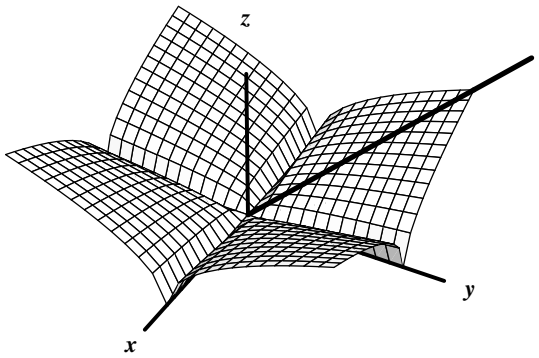


Figure 12.4. Graph of $f(x,y) = \sqrt{|xy|}$ which is continuous and nondifferentiable at $(0,0)$.

tion of Definition 12.13 that we have just given for the differentiability of a function of two variables.

Definition 12.16: Let f be defined on an open set $D \subset \mathbb{R}^n$ and let

$$\mathbf{x} = (x_1, \ldots, x_n) \in D.$$

Then f is *differentiable* at $\mathbf{x}$ if

1. The partial derivatives $f_1, \ldots, f_n$ all exist (finite) at $\mathbf{x}$, and
2. When $\varepsilon = \varepsilon(h_1, \ldots, h_n)$ is defined by $\varepsilon(0, \ldots, 0) = 0$ and
$$f(x_1 + h_1, \ldots, x_n + h_n) - f(x_1, \ldots, x_n) = \sum_{i=1}^{n} f_i(x_1, \ldots, x_n)h_i + \varepsilon\sqrt{h_1^2 + \cdots + h_n^2},$$
we have $\varepsilon(h_1, \ldots, h_n) \to 0$ when $\sqrt{h_1^2 + \cdots + h_n^2} \to 0$.

If f is differentiable at all points in D, we say f is *differentiable* on D.

Note. Statement (2) in the definition is given using the usual euclidean norm. As we have seen before in the note following Definition 12.13, that gives the $\mathbb{R}^2$ case, it often simplifies proofs involving differentiability to use the $\|\cdot\|_1$ norm. Exercise 12.4.2 shows that this is possible. Using the $\|\cdot\|_1$ norm, then statement (2) becomes

$2'$. When $\varepsilon = \varepsilon(h_1, \ldots, h_n)$ is defined by
$$f(x_1 + h_1, \ldots, x_n + h_n) - f(x_1, \ldots, x_n) = \sum_{i=1}^{n} f_i(x_1, \ldots, x_n)h_i + \varepsilon(|h_1| + \cdots + |h_n|)$$
with $\varepsilon(0, \ldots, 0) = 0$, we have $\varepsilon \to 0$ when $|h_1| + \cdots + |h_n| \to 0$.

See also Exercise 12.4.3.